Answer Key & Explanations — Quiz/Practice me add kiye gaye Questions

Verification Copy — 197 self-authored questions (waves 1–5) · Har question: stem (EN/HI) + 4 options + ✓ **correct answer** + why (EN/HI) + nukte + source · Ye un questions ka key hai jo source banks se NAHI aaye (topic books me inke answers nahi the) · Compiled by PRAKASH • EkGuru Learning · Prepared for: RPSC Programmer • ACP • RSMSSB IA • SA • DD • BCI • SCI • REET • RAS • DSSSB • KVS • NVS • HTET • SSC CGL/CHSL • RRB • IBPS • UPSC & all Rajasthan + Central Govt. Exams (Computer Awareness)

Part A — Wave-5 NEW (new/ material ke gap concepts) — 26 Q

Topic 1 — Data Communications & Networks

Q1. In a _____ network, the communication channel is shared by all the machines on the network:

_____ network में communication channel network की सभी machines में साझा होता है:

✓ **(A) Broadcast (multipoint) network**
(B) Point-to-point network
(C) Circuit-switched WAN
(D) Extranet

✓ **Answer: (A) Broadcast (multipoint) network**

Why: Broadcast/multipoint network mein ek shared channel sab hosts tak pahunchta hai (jaise bus/Ethernet) — point-to-point mein sirf do ends hote hain.

क्यों: Broadcast/multipoint network में एक shared channel सभी hosts तक जाता है; point-to-point में सिर्फ दो ends।

Nukte: Yaad rakho: point-to-point = dedicated link; multipoint/broadcast = shared medium.

[Source: Practice — original (textbook-based)] · Medium

Q2. In _____ switching, the resources needed for communication are reserved for the duration of the session:

_____ switching में, संचार के लिए आवश्यक संसाधन पूरे session की अवधि के लिए आरक्षित रहते हैं:

✓ **(A) Circuit**
(B) Packet
(C) Message
(D) Cell

✓ **Answer: (A) Circuit**

Why: Circuit switching: connection ke pehle path + bandwidth reserve hoti hai (telephone jaisa); packet switching mein koi reservation nahi.

क्यों: Circuit switching में path + bandwidth पहले से आरक्षित; packet switching में कोई reservation नहीं।

Nukte: Reservation = circuit; store-and-forward = packet/message.

[Source: Practice — original (textbook-based)] · Medium

Topic 2 — OSI & TCP/IP Models

Q3. The list of protocols used by a system — one protocol per layer — is called a:
किसी system द्वारा प्रयोग की जाने वाली protocols की सूची — हर layer पर एक protocol — कहलाती है:

✓ **(A) Protocol stack (protocol suite)**
(B) Protocol syntax
(C) Encapsulation table
(D) Address map

✓ **Answer: (A) Protocol stack (protocol suite)**

Why: Protocol stack = layered list (jaise TCP/IP stack: IP, TCP, HTTP...). Har layer apna protocol chunta hai — 'suite' bhi kehte hain.

क्यों**:** Protocol stack = layer-wise सूची (जैसे TCP/IP stack)।

Nukte: Syntax = message ka format; stack = layers ka protocol set.

[Source: Practice — original (textbook-based)] · Medium

Topic 3 — Internetworking & Addressing

Q4. In IPv6, the field that determines the lifetime of a datagram (the counterpart of IPv4's TTL) is called:
IPv6 में, datagram का जीवन-काल तय करने वाला field (IPv4 के TTL के समकक्ष) कहलाता है:

✓ **(A) Hop Limit**
(B) Flow Label
(C) Next Header
(D) Payload Length

✓ **Answer: (A) Hop Limit**

Why: IPv6 Hop Limit = har hop par 1 ghat-ta hai (TTL jaisa); Flow Label = QoS flow pehchaan; Next Header = agla extension/upper-layer header; Payload Length = data ki lambai.

क्यों**:** IPv6 Hop Limit = हर hop पर 1 घटता है (TTL जैसा)। बाकी fields अन्य कामों के लिए।

Nukte: IPv6 ke 8 base-header fields: Version, Traffic Class, Flow Label, Payload Length, Next Header, Hop Limit, Src, Dst.

[Source: Practice — original (textbook-based)] · Medium

Q5. In the dual-stack approach to IPv6 migration, a host or router runs:
IPv6 migration के dual-stack तरीके में, host या router चलाता है:

✓ **(A) Both IPv4 and IPv6 protocols simultaneously** / IPv4 और IPv6 दोनों protocols साथ-साथ
(B) Only IPv6 with IPv4 tunnels / सिर्फ IPv6, IPv4 tunnels के साथ
(C) Only IPv4 with IPv6 headers / सिर्फ IPv4, IPv6 headers के साथ
(D) Neither — only translation / कोई नहीं — सिर्फ translation

✓ **Answer: (A) Both IPv4 and IPv6 protocols simultaneously**

Why: Dual stack: dono protocols ek saath — jo destination IPv4 samajhe woh IPv4 se, jo IPv6 woh IPv6 se. Tunneling/translation alag strategies hain.

क्यों**:** Dual stack: दोनों protocols साथ; tunneling/translation अलग रणनीतियाँ हैं।

Nukte: IPv6 migration ke 3 tarika: dual stack, tunneling, translation.

[Source: Practice — original (textbook-based)] · Hard

Q6. Teredo is an IPv6 transition technique that provides IPv6 connectivity to hosts behind NAT by:

Teredo एक IPv6 transition तकनीक है जो NAT के पीछे के hosts को IPv6 connectivity देती है:

✓ **(A) Encapsulating IPv6 packets inside UDP datagrams over IPv4** / IPv6 packets को IPv4 पर UDP datagrams के अंदर encapsulate करके

(B) Translating every IPv6 header to IPv4 / हर IPv6 header को IPv4 में बदलकर

(C) Reserving public IPv6 blocks per host / हर host के लिए सार्वजनिक IPv6 blocks आरक्षित करके

(D) Broadcasting IPv6 router solicitations / IPv6 router solicitations broadcast करके

✓ **Answer: (A) Encapsulating IPv6 packets inside UDP datagrams over IPv4**

Why: Teredo (RFC 4380): IPv6-in-UDP-over-IPv4 tunneling — NAT traversal ke liye UDP ports use hote hain. 6to4 alag hai (IPv6-in-IPv4, proto 41).

क्यों: Teredo (RFC 4380): IPv6-in-UDP-over-IPv4 tunneling — NAT traversal हेतु UDP ports।

Nukte: Teredo = UDP; 6to4 = IPv4 protocol 41; dual stack = dono protocols native.

[Source: Practice — original (textbook-based)] · Medium

Topic 5 — Telephony, TV, Cable & Wireless

Q7. A coaxial cable consists of _____ concentric copper conductors:

Coaxial cable में _____ संकेंद्रित (concentric) copper conductors होते हैं:

✓ **(A) Two (inner wire + outer shield)** / दो (भीतरी तार + बाहरी shield)

(B) Four / चार

(C) Eight / आठ

(D) One / एक

✓ **Answer: (A) Two (inner wire + outer shield)**

Why: Coax: beech mein inner copper conductor, uske chaaron taraf outer copper shield — dono ke beech insulator. Isi se EMI resistance aati hai.

क्यों: Coax: भीतरी copper conductor और बाहरी copper shield — बीच में insulator।

Nukte: UTP = 4 pairs (8 wires); coax = 2 concentric conductors.

[Source: Practice — original (textbook-based)] · Medium

Q8. 10Base5, 10Base2 and 10BaseT are different types of:

10Base5, 10Base2 और 10BaseT हैं किसके अलग-अलग प्रकार:

✓ **(A) Ethernet**

(B) Token Ring

(C) FDDI

(D) ATM

✓ **Answer: (A) Ethernet**

Why: Ye sab Ethernet cable standards hain: 10 Mbps; Base = baseband. 10Base5 = thick coax (Thicknet), 10Base2 = thin coax (Thinnet), 10BaseT = UTP.

क्यों: ये सब Ethernet cable standards हैं (10 Mbps, baseband): 10Base5 thick coax, 10Base2 thin coax, 10BaseT UTP।

Nukte: Suffix: 5/2 = segment length (sau metre mein); T = twisted pair.

[Source: Practice — original (textbook-based)] · Medium

Q9. In the Ethernet names 10Base2 and 10Base5, the digits '2' and '5' indicate:
Ethernet नामों 10Base2 और 10Base5 में अंक '2' और '5' बताते हैं:

✓ **(A) Maximum segment length in hundreds of metres (~200 m and ~500 m)** / अधिकतम segment लंबाई, सौ मीटर में (~200 m और ~500 m)
(B) Number of hosts per segment / प्रति segment hosts की संख्या
(C) Cable thickness in mm / Cable की मोटाई mm में
(D) Transmission speed in Mbps / Transmission गति Mbps में

✓ **Answer: (A) Maximum segment length in hundreds of metres (~200 m and ~500 m)**

Why: 10Base2 = ~185-200 m (thin coax), 10Base5 = 500 m (thick coax) — '10' = 10 Mbps, 'Base' = baseband.

क्यों**:** 10Base2 = ~185-200 m; 10Base5 = 500 m; '10' = 10 Mbps; 'Base' = baseband।

Nukte: Speed '10' mein hai; digits 2/5 distance mein (sau metre).

[Source: Practice — original (textbook-based)] · Medium

Q10. The function of a DSLAM (DSL Access Multiplexer) located at the telephone exchange is to:
Telephone exchange पर स्थित DSLAM (DSL Access Multiplexer) का काम है:

✓ **(A) Aggregate many DSL subscriber lines onto a high-speed backbone link** / कई DSL subscriber lines को high-speed backbone link पर इकट्ठा करना
(B) Convert digital voice to analog / Digital voice को analog में बदलना
(C) Assign IP addresses to DSL modems / DSL modems को IP addresses देना
(D) Encrypt DSL traffic / DSL traffic encrypt करना

✓ **Answer: (A) Aggregate many DSL subscriber lines onto a high-speed backbone link**

Why: DSLAM: exchange par sab DSL lines multiplex karke backbone par daalta hai (aur aksar ISP routing tak). Ghar par DSL modem hota hai.

क्यों**:** DSLAM: exchange पर सभी DSL lines को backbone पर multiplex करता है; घर पर DSL modem होता है।

Nukte: DHCP = IP dena; modem = signal convert; DSLAM = aggregation.

[Source: Practice — original (textbook-based)] · Medium

Q11. Cable Internet access (HFC) differs from DSL in that:
Cable Internet access (HFC), DSL से भिन्न है क्योंकि:

✓ **(A) The HFC bandwidth is shared among all subscribers on the neighbourhood segment** / HFC bandwidth पड़ोस के segment के सभी subscribers में साझा होती है
(B) It uses the telephone line's DSLAM / यह telephone line का DSLAM प्रयोग करता है
(C) It cannot carry television signals simultaneously / यह television signals साथ नहीं चला सकता
(D) Each subscriber gets a dedicated fibre pair / हर subscriber को dedicated fibre जोड़ी मिलती है

✓ **Answer: (A) The HFC bandwidth is shared among all subscribers on the neighbourhood segment**

Why: HFC (Hybrid Fibre-Coax): fibre tak, phir coax — segment ka bandwidth shared (peak hours par slow). DSL mein har line exchange tak dedicated hoti hai.

क्यों**:** HFC: bandwidth साझा (peak hours पर धीमा); DSL में हर line exchange तक dedicated।

Nukte: DSL = dedicated last-mile; Cable = shared coax segment.

[Source: Practice — original (textbook-based)] · Medium

Q12. In SONET (Synchronous Optical Networking), the base electrical signal level STS-1 has a data rate of approximately:

SONET में, आधार electrical signal level STS-1 की data rate लगभग होती है:

✓ **(A) 51.84 Mbps**
(B) 155.52 Mbps
(C) 622.08 Mbps
(D) 43.2 Mbps

✓ **Answer: (A) 51.84 Mbps**

Why: STS-1 = 51.84 Mbps; STS-3 = 155.52 (OC-3); STS-12 = 622.08 (OC-12) — har level n × STS-1.

क्यों**:** STS-1 = 51.84 Mbps; STS-3 = 155.52; STS-12 = 622.08 — हर level n × STS-1।

Nukte: Yaad karo: STS-1 = 51.84 Mbps, aur OC-n/STS-n = n × 51.84.

[Source: Practice — original (textbook-based)] · Hard

Q13. In the SONET protocol stack, the photonic layer corresponds to which layer of the OSI model?

SONET protocol stack में, photonic layer OSI model की किस layer के बराबर है?

✓ **(A) Physical layer**
(B) Data link layer
(C) Network layer
(D) Session layer

✓ **Answer: (A) Physical layer**

Why: SONET ke 4 layers: Photonic (physical/electrical-optical), Section, Line, Path — Photonic layer light signals/green-red lasers ka kaam sambhalti hai.

क्यों**:** SONET की 4 layers: Photonic, Section, Line, Path — Photonic = physical।

Nukte: Section/Line/Path = overhead management levels; Photonic = actual light.

[Source: Practice — original (textbook-based)] · Medium

Q14. Terrestrial radio channels are broadly classified into _____ groups:

Terrestrial radio channels मोटे तौर पर _____ समूहों में विभाजित हैं:

✓ **(A) Three** / तीन
(B) Two / दो
(C) Five / पाँच
(D) Seven / सात

✓ **Answer: (A) Three**

Why: Forouzan: terrestrial radio = 3 groups — (1) VLF/LF/MF (ground waves), (2) HF (sky waves/ionosphere reflection), (3) VHF/UHF/SHF/EHF (line-of-sight).

क्यों**:** Forouzan: terrestrial radio = 3 समूह — ground wave, sky wave, line-of-sight।

Nukte: Satellite/microwave alag category hain; terrestrial = teen.

[Source: Practice — original (textbook-based)] · Medium

Topic 6 — Transport, Routing & Application

Q15. Network congestion occurs when:

Network congestion तब होता है जब:

✓ **(A) The offered traffic exceeds the network's capacity** / Offered traffic network की क्षमता से अधिक हो

(B) Two hosts use the same port / दो hosts एक ही port प्रयोग करें

(C) A frame is corrupted by noise / Noise से frame खराब हो

(D) The DNS server is down / DNS server बंद हो

✓ **Answer: (A) The offered traffic exceeds the network's capacity**

Why: Congestion = queues bhar jaate hain kyunki demand capacity se zyada hai — delay/loss badhte hain. Isi ko sambhalne ke liye congestion control hota hai.

क्यों**:** Congestion = demand क्षमता से अधिक; queues भरती हैं, delay/loss बढ़ता है।

Nukte: Flow control = receiver ki speed; congestion = network ki capacity.

[Source: Practice — original (textbook-based)] · Medium

Q16. If the Protocol field of an IPv4 header is 17, the transport-layer protocol carried in the datagram is:

यदि IPv4 header का Protocol field 17 हो, तो datagram में लिया गया transport-layer protocol है:

✓ **(A) UDP**

(B) TCP

(C) ICMP

(D) IGMP

✓ **Answer: (A) UDP**

Why: Protocol numbers: 1 = ICMP, 2 = IGMP, 6 = TCP, 17 = UDP, 89 = OSPF. Ye field fragmentation ke saath-saath next-header ki pehchaan deta hai.

क्यों**:** Protocol numbers: 1=ICMP, 2=IGMP, 6=TCP, 17=UDP, 89=OSPF।

Nukte: 6 TCP, 17 UDP — dono sabse zyada puche jaate hain.

[Source: Practice — original (textbook-based)] · Medium

Q17. In IPv4 fragmentation, the LAST fragment has the More Fragments (MF) flag set to:

IPv4 fragmentation में, अंतिम fragment का More Fragments (MF) flag होता है:

✓ **(A) 0**

(B) 1

(C) Its fragment offset value / उसका fragment offset मान

(D) The same as the first fragment / पहले fragment के बराबर

✓ **Answer: (A) 0**

Why: MF = 1 matlab 'aur fragments aane hain'; aakhri fragment par MF = 0. Har fragment ka offset alag hota hai.

क्यों**:** MF=1 = 'और fragments आने हैं'; अंतिम पर MF=0।

Nukte: Dono fields (MF flag + offset) milkar reassembly karte hain.

[Source: Practice — original (textbook-based)] · Hard

Q18. The Fragment Offset field in the IPv4 header is used to:

IPv4 header का Fragment Offset field प्रयोग होता है:

✓ **(A) Place the fragment at its correct position in the original datagram during reassembly** / Reassembly के दौरान fragment को मूल datagram में सही स्थान पर रखने के लिए

(B) Count the total number of fragments / कुल fragments गिनने के लिए

(C) Encrypt fragment payloads / Fragment payloads encrypt करने के लिए

(D) Set the fragment's lifetime / Fragment का जीवन-काल तय करने के लिए

✓ Answer: (A) Place the fragment at its correct position in the original datagram during reassembly

Why: Offset (8-byte units mein) batata hai fragment original datagram mein kahan baitha hai — destination sirf offset se hi sahi order bana sakta hai.

क्यों**:** Offset (8-byte units में) बताता है fragment का स्थान — इसी से सही क्रम बनता है।

Nukte: Offset = position; MF = aur aane hain ya nahi.

[Source: Practice — original (textbook-based)] · Hard

Q19. The data field of an IPv4 datagram can carry:

IPv4 datagram का data field ले जा सकता है:

✓ **(A) TCP segment, UDP datagram, or ICMP message — all of these** / TCP segment, UDP datagram, या ICMP message — ये सब

(B) Only TCP segments / सिर्फ TCP segments

(C) Only UDP datagrams / सिर्फ UDP datagrams

(D) Only application data / सिर्फ application data

✓ Answer: (A) TCP segment, UDP datagram, or ICMP message — all of these

Why: IP ka payload kuch bhi ho sakta hai jo Protocol field bataye — TCP/UDP segments, ICMP/IGMP messages, tunnels mein doosra IP packet bhi.

क्यों**:** IP का payload कुछ भी हो सकता है — TCP/UDP, ICMP/IGMP, या tunnel में दूसरा IP packet।

Nukte: Protocol field = 6/17/1/2/41... isi se pata chalta hai kya andar hai.

[Source: Practice — original (textbook-based)] · Hard

Q20. RSVP (Resource ReSerVation Protocol) is used by hosts to:

RSVP (Resource ReSerVation Protocol) का उपयोग hosts करते हैं:

✓ **(A) Request reservation of network resources (bandwidth) along a path for QoS flows** / QoS flows के लिए रास्ते में network संसाधन (bandwidth) आरक्षित करने के अनुरोध हेतु

(B) Assign IP addresses automatically / स्वतः IP addresses देने के लिए

(C) Resolve domain names / Domain names हल करने के लिए

(D) Exchange routing tables between routers / Routers के बीच routing tables बाँटने के लिए

✓ Answer: (A) Request reservation of network resources (bandwidth) along a path for QoS flows

Why: RSVP = signalling protocol jo routers se kehta hai ki flow ke liye resources reserve karo (IntServ/QoS). IP assignment = DHCP; routing = OSPF/BGP; names = DNS.

क्यों**:** RSVP = routers से संसाधन आरक्षित करने का संकेत (QoS)। IP=DHCP; routing=OSPF/BGP; names=DNS।

Nukte: RSVP 'reservation' ka protocol hai — circuit-switching jaisi guarantee nahi, bas soft-state reservations.

[Source: Practice — original (textbook-based)] · Medium

Q21. The first line of an HTTP request message is called the:

HTTP request message की पहली पंक्ति कहलाती है:

✓ **(A) Request line**
(B) Header line
(C) Status line
(D) Body line

✓ **Answer: (A) Request line**

Why: Request line = METHOD URI HTTP/VERSION (jaise 'GET /index.html HTTP/1.1'). Response message ki pehli line STATUS line hoti hai.

क्यों**:** Request line = METHOD URI HTTP/VERSION। Response की पहली line = status line।

Nukte: Status line sirf RESPONSE mein hoti hai.

[Source: Practice — original (textbook-based)] · Medium

Q22. By default, HTTP/1.1 connections are:

डिफ़ॉल्ट रूप से, HTTP/1.1 connections होते हैं:

✓ **(A) Persistent (keep-alive)**
(B) Non-persistent
(C) Simplex
(D) Datagram-based

✓ **Answer: (A) Persistent (keep-alive)**

Why: HTTP/1.1 default persistent connection: ek TCP connection par kai request/response (Connection: close se band hota hai). HTTP/1.0 default non-persistent tha.

क्यों**:** HTTP/1.1 default persistent: एक connection पर कई request/response; HTTP/1.0 non-persistent था।

Nukte: Persistent = connection reuse; non-persistent = har object par nayi connection.

[Source: Practice — original (textbook-based)] · Medium

Q23. In HTTP, a conditional GET (using If-Modified-Since) allows:

HTTP में conditional GET (If-Modified-Since के साथ) संभव बनाता है:

✓ **(A) Cache validation — the server sends a 304 Not Modified instead of the object if unchanged** / Cache validation — अपरिवर्तित होने पर server वस्तु के बजाय 304 Not Modified भेजे
(B) Automatic language translation / स्वतः भाषा अनुवाद
(C) Password-free login / बिना password login
(D) Parallel downloads / समांतर downloads

✓ **Answer: (A) Cache validation — the server sends a 304 Not Modified instead of the object if unchanged**

Why: Conditional GET: browser poochta hai 'kitne din se badla?'; agar object same hai to sirf 304 header — bandwidth bachti hai.

क्यों**:** Conditional GET: object अपरिवर्तित हो तो सिर्फ 304 Not Modified — bandwidth बचती है।

Nukte: 304 = Not Modified; 200 = OK (poora content).

[Source: Practice — original (textbook-based)] · Medium

Q24. Which of the following is NOT a valid HTTP status code?
निम्न में से कौन-सा वैध HTTP status code नहीं है?

✓ **(A) 250**
(B) 200
(C) 404
(D) 503

✓ **Answer: (A) 250**

Why: 200 = OK, 3xx redirection (301/302/304), 4xx client errors (400/403/404), 5xx server errors (500/503). '250' SMTP ka 'mail action ok' code hai, HTTP ka nahi.

क्यों**:** 200=OK; 3xx=redirection; 4xx=client error; 5xx=server error। 250 SMTP का code है।

Nukte: HTTP 2xx family: 200, 201, 204... 250 nahi.

[Source: Practice — original (textbook-based)] · Hard

Topic 7 — Network Security

Q25. A honeypot in network security is:
Network security में honeypot होता है:

✓ **(A) A decoy system set up to attract attackers and study their activity** / एक प्रलोभन (decoy) system जो attackers को आकर्षित कर उनकी गतिविधि अध्ययन करता है
(B) A firewall rule that blocks all traffic / सभी traffic रोकने वाला firewall rule
(C) A backup server / एक backup server
(D) An antivirus quarantine folder / Antivirus quarantine folder

✓ **Answer: (A) A decoy system set up to attract attackers and study their activity**

Why: Honeypot = jaal jaisa fake system/data — attackers usme fass kar apne tools/tarike dikha dete hain; defenders ko intelligence milti hai.

क्यों**:** Honeypot = जाल जैसा decoy system — attackers अपने तरीके दिखा देते हैं।

Nukte: Firewall rokta hai; honeypot fasata hai aur seekhata hai.

[Source: Practice — original (textbook-based)] · Medium

Q26. A shadow honeypot combines a honeypot with:
Shadow honeypot जोड़ता है honeypot के साथ:

✓ **(A) An anomaly detection system (suspect traffic is run inside the honeypot)** / Anomaly detection system (संदिग्ध traffic को honeypot के अंदर चलाया जाता है)
(B) A physical lock / एक भौतिक lock
(C) A DNS sinkhole only / सिर्फ DNS sinkhole
(D) A load balancer / एक load balancer

✓ **Answer: (A) An anomaly detection system (suspect traffic is run inside the honeypot)**

Why: Shadow honeypot: IDS/IPS ka anomaly detector jo kuch block kare, uske shadow copy ko honeypot par chala kar dekhte hain ki attack tha ya false positive — learning + safety dono.

क्यों**:** Shadow honeypot: anomaly detector की shadow copy को honeypot पर चलाकर जाँच — attack या false positive।

Nukte: Simple honeypot = decoy; shadow = decoy + anomaly detector ka validation.

[Source: Practice — original (textbook-based)] · Medium

Part B — Waves 1–4 originals (pehle se bank me) — 171 Q

Topic 1 — Data Communications & Networks

Q1. In a traditional ring network, repeaters are particularly associated with which topology because each station can regenerate and forward the signal?
पारंपरिक ring network में repeaters किस topology से विशेष रूप से जुड़े हैं, क्योंकि हर station signal को regenerate और आगे भेज सकता है?

✓ **(A) Ring topology**
(B) Mesh topology
(C) Bus topology
(D) Star topology

✓ **Answer: (A) Ring topology**

Why: In a classic/traditional ring, each station commonly regenerates (repeats) the incoming signal before passing it to the next station — repeaters are thus built into the ring model.
क्यों: पारंपरिक ring में हर station आने वाले signal को अगले station को देने से पहले regenerate (repeat) करता है — इसलिए ring model में repeaters अंतर्निहित हैं।

Nukte: Concept-check correction of the old "repeater is used in ___ topology" stem (that wording wrongly implied repeaters are exclusive to one topology). Repeaters are Physical-layer devices usable on many links; the "traditional ring network" qualifier keeps the intended textbook answer (Ring) without the false universal claim. Not a verified PYQ.

[Source: Practice — original (concept-check correction — not a verified PYQ)] · Medium

Q2. Which of the following is NOT a component of data communication?
निम्न में से कौन data communication का component नहीं है?

(A) Message
(B) Protocol
✓ **(C) Compiler**
(D) Transmission medium

✓ **Answer: (C) Compiler**

Why: Message, sender, receiver, medium, protocol — paanch components; compiler programming ka tool hai.
क्यों: पाँच components; compiler programming का tool है।

Nukte: Odd-one-out: compiler ka data communication se koi sambandh nahi.

[Source: Practice — original (textbook-based)] · Hard

Q3. In a protocol, the meaning of each section of the message is called:
एक protocol में message के हर भाग के अर्थ को क्या कहते हैं?

(A) Syntax
✓ **(B) Semantics**
(C) Timing
(D) Encoding

✓ **Answer: (B) Semantics**

Why: Semantics = har field/section ka meaning; syntax = format; timing = speed/order.
क्यों: Semantics = हर भाग का अर्थ; syntax = format; timing = क्रम।

Nukte: Syntax format dikhata hai, semantics matlab.

[Source: Practice — original (textbook-based)] · Medium

Q4. Coaxial cable and fiber optic cable are examples of:

Coaxial cable और fiber optic cable किसके उदाहरण हैं?

✓ **(A) Guided media**
(B) Unguided media
(C) Wireless media
(D) Simplex media

✓ **Answer: (A) Guided media**

Why: Dono wired (guided) media hain — signal ek physical medium se guzarta hai.

क्यों**:** दोनों wired (guided) media हैं।

Nukte: Wireless = unguided (radio, microwave).

[Source: Practice — original (textbook-based)] · Medium

Q5. The three aspects of TIMING in a protocol are speed matching, sequencing, and:

एक protocol में TIMING के तीन पक्ष हैं — speed matching, sequencing और:

✓ **(A) Flow control**
(B) Error control
(C) Syntax
(D) Addressing

✓ **Answer: (A) Flow control**

Why: Forouzan: protocol ke 3 elements — syntax (format), semantics (meaning), timing (speed matching + sequencing + flow control).

क्यों**:** protocol के 3 elements — syntax, semantics, timing (speed matching + sequencing + flow control)।

Nukte: Yaad rakho: timing = events ka order + speed + flow control; error control reliability ke liye alag concept hai.

[Source: Practice — original (textbook-based)] · Hard

Q6. If packets arrive at the receiver with uneven delay, this problem is called:

यदि packets receiver पर असमान delay से पहुँचें, तो इस समस्या को क्या कहते हैं?

(A) Attenuation
✓ **(B) Jitter**
(C) Distortion
(D) Noise

✓ **Answer: (B) Jitter**

Why: Uneven/variable delay = jitter; audio-video playback kharab karta hai.

क्यों**:** असमान delay = jitter।

Nukte: Attenuation signal weakness hai; jitter delay ka variation.

[Source: Practice — original (textbook-based)] · Medium

Q7. The amount of data that is actually delivered successfully per unit time is called:

प्रति इकाई समय में सफलतापूर्वक पहुँचा data कहलाता है:

(A) Bandwidth
(B) Latency
✓ **(C) Throughput**
(D) Jitter

✓ **Answer: (C) Throughput**

Why: Throughput = asli (effective) data transfer rate; bandwidth = maximum capacity.

क्यों**:** Throughput = वास्तविक data rate; bandwidth = अधिकतम क्षमता।

Nukte: Bandwidth nominal capacity hai, throughput actual.

[Source: Practice — original (textbook-based)] · Medium

Q8. MTBF (Mean Time Between Failures) is related to which criteria of a network?
MTBF network के किस criteria से संबंधित है?

(A) Performance
✓ **(B) Reliability**
(C) Security
(D) Accuracy

✓ **Answer: (B) Reliability**

Why: Reliability failures ki frequency aur recovery se naapi jaati hai — MTBF isi ka measure hai.
क्यों**:** Reliability = विफलता दर + ठीक होना; MTBF उसी का माप।
Nukte: Performance = throughput/latency.
[Source: Practice — original (textbook-based)] · Medium

Q9. Which of the following is a measure of network RELIABILITY (not performance)?
निम्न में से कौन network की RELIABILITY (performance नहीं) का माप है?

(A) Throughput
✓ **(B) Frequency of failure** / Failure की आवृत्ति
(C) Jitter
(D) Delay

✓ **Answer: (B) Frequency of failure**

Why: Forouzan: reliability = failure ki frequency + recovery time; throughput, delay, jitter — sab performance criteria hain.
क्यों**:** reliability = failure की आवृत्ति + recovery time; throughput, delay, jitter — performance criteria हैं।
Nukte: Performance = speed/quality of delivery; Reliability = kitni baar network toota aur kitni jaldi theek hua.
[Source: Practice — original (textbook-based)] · Hard

Q10. In which mode can both stations transmit and receive at the same instant?
किस mode में दोनों stations एक ही समय में भेज और पा सकते हैं?

(A) Simplex
(B) Half-duplex
✓ **(C) Full-duplex**
(D) Broadcast

✓ **Answer: (C) Full-duplex**

Why: Full-duplex = dono taraf ek saath (telephone).
क्यों**:** Full-duplex = दोनों तरफ़ एक साथ।
Nukte: Half-duplex mein baari lagti hai.
[Source: Practice — original (textbook-based)] · Hard

Q11. Which of the following works in simplex mode?
निम्न में से कौन simplex mode में काम करता है?

(A) Telephone
(B) Walkie-talkie
✓ **(C) Keyboard and monitor** / Keyboard और monitor
(D) Chat application

✓ **Answer: (C) Keyboard and monitor**

Why: Keyboard sirf bhejta hai, monitor sirf dikhata hai — ek hi direction.
क्यों**:** Keyboard सिर्फ़ भेजता है, monitor सिर्फ़ दिखाता है।
Nukte: Telephone = full; walkie-talkie = half.
[Source: Practice — original (textbook-based)] · Medium

Q12. In 100BASE-TX (Fast Ethernet over UTP), full-duplex operation is possible because:
100BASE-TX (UTP पर Fast Ethernet) में full-duplex संभव है क्योंकि:

✓ **(A) UTP uses separate wire pairs for transmitting and receiving** / UTP में transmit और receive के लिए अलग wire pairs हैं
(B) UTP uses time-division multiplexing / UTP में time-division multiplexing है
(C) UTP uses two different frequencies / UTP में दो अलग frequencies हैं
(D) Two UTP cables are used / दो UTP cables प्रयोग होती हैं

✓ **Answer: (A) UTP uses separate wire pairs for transmitting and receiving**
Why: 100BASE-TX pairs 1-2 aur 3-6 — ek pair TX, doosra RX; isliye dono directions ek saath (full-duplex) chal sakte hain.
क्यों**:** pairs 1-2 और 3-6 — एक pair TX, दूसरा RX; दोनों दिशाएँ साथ चलती हैं।
Nukte: Half-duplex mein ek hi pair dono directions ke liye hota (collision possible); alag pairs se collision hi nahi.
[Source: Practice — original (textbook-based)] · Medium

Q13. In a mesh topology with 6 devices, how many dedicated links are required?
6 devices की mesh topology में कितने dedicated links चाहिए?

✓ **(A) 15**
(B) 30
(C) 36
(D) 12

✓ **Answer: (A) 15**
Why: $n(n-1)/2 = 6\times5/2 = 15$ links.
क्यों**:** $n(n-1)/2 = 6\times5/2 = 15$।
Nukte: $30 = n(n-1)$ full duplex links counting ulta.
[Source: Practice — original (textbook-based)] · Hard

Q14. If the central device of a star topology fails, the result is:
यदि star topology का central device ख़राब हो जाए तो:

(A) Only one device stops / केवल एक device रुकता है
✓ **(B) Whole network stops** / पूरा network रुक जाता है
(C) Network works as mesh / Network mesh जैसा चलता है
(D) Only data transfer slows / केवल data transfer धीमा होता है

✓ **Answer: (B) Whole network stops**
Why: Star mein central hub/switch single point of failure hai.
क्यों**:** Star में central device = single point of failure।
Nukte: Mesh mein ek link fail = sirf ek device.
[Source: Practice — original (textbook-based)] · Medium

Q15. Which topology gives every device equal access using a token?
कौन-सी topology token द्वारा हर device को बराबर पहुँच देती है?

(A) Bus
(B) Star
✓ **(C) Ring**
(D) Mesh

✓ **Answer: (C) Ring**
Why: Ring (Token Ring) mein token pass hokar equal access deta hai.
क्यों**:** Ring (Token Ring) में token से बराबर पहुँच।
Nukte: Bus/Star mein token nahi chalta.
[Source: Practice — original (textbook-based)] · Medium

Q16. A hybrid topology is:

Hybrid topology है:

(A) Only mesh / केवल mesh
✓ **(B) Combination of two or more topologies** / दो या अधिक topologies का मिश्रण
(C) Only bus / केवल bus
(D) A wireless topology / एक wireless topology

✓ **Answer: (B) Combination of two or more topologies**

Why: Hybrid = do ya zyada basic topologies ka combination (star-of-stars).

क्यों**:** Hybrid = दो या अधिक topologies का मिश्रण।

Nukte: Wireless topology ka term nahi.

[Source: Practice — original (textbook-based)] · Medium

Q17. In a bus topology, both ends of the cable are terminated with a resistor to prevent:

Bus topology में cable के दोनों सिरों पर resistor लगाया जाता है ताकि रोका जा सके:

✓ **(A) Signal reflection (bounce)**
(B) Crosstalk
(C) Attenuation
(D) Dispersion

✓ **Answer: (A) Signal reflection (bounce)**

Why: Bina termination ke travelling signal end par reflect hokar wapas aata hai aur garbage create karta hai — terminator 50-ohm resistor ye bounce khatam karta hai.

क्यों**:** bina termination के signal reflect होकर garbage बनाता है — 50-ohm terminator रोकता है।

Nukte: Terminator = signal ko absorb karne wala resistor; crosstalk = adjacent wires ka interference.

[Source: Practice — original (textbook-based)] · Hard

Q18. A company's internal network that uses Internet technology but is accessible only to its employees is called:

Internet technology पर चलने वाला वह internal network जो केवल कर्मचारियों के लिए हो:

(A) Internet
(B) Extranet
✓ **(C) Intranet**
(D) WAN

✓ **Answer: (C) Intranet**

Why: Intranet = organization ka private network (Internet jaisa technology).

क्यों**:** Intranet = संस्था का private network।

Nukte: Extranet = authorized bahari users bhi.

[Source: Practice — original (textbook-based)] · Medium

Q19. In a peer-to-peer network:

Peer-to-peer network में:

(A) One computer is always the server / एक computer हमेशा server होता है

✓ **(B) Every computer can act as client and server** / हर computer client और server दोनों बन सकता है

(C) Only files can be shared / केवल files साझा होती हैं

(D) It cannot work without a hub / Hub के बिना नहीं चलता

✓ **Answer: (B) Every computer can act as client and server**

Why: P2P mein koi dedicated server nahi — har computer dono bhumika nibhata hai.

क्यों**:** P2P में dedicated server नहीं — हर computer दोनों भूमिका में।

Nukte: Client-server mein fixed server hota hai.

[Source: Practice — original (textbook-based)] · Medium

Q20. When several senders share one communication line through a common path, the access is called:

जब कई senders एक ही line साझा करते हैं, तो वह access है:

(A) Point-to-point

(B) Dedicated

✓ **(C) Multipoint**

(D) Simplex

✓ **Answer: (C) Multipoint**

Why: Multipoint (multi-drop) mein ek link par multiple nodes share karte hain.

क्यों**:** Multipoint में एक link पर कई nodes साझा करते हैं।

Nukte: Point-to-point = sirf do nodes.

[Source: Practice — original (textbook-based)] · Medium

Q21. A high-speed network that connects servers to storage devices (disk arrays, tape libraries) is called:

Servers को storage devices (disk arrays, tape libraries) से जोड़ने वाली high-speed network कहलाती है:

✓ **(A) SAN (Storage Area Network)**

(B) CAN (Campus Area Network)

(C) PAN (Personal Area Network)

(D) MAN (Metropolitan Area Network)

✓ **Answer: (A) SAN (Storage Area Network)**

Why: SAN data-centre technology hai — servers aur storage ko dedicated high-speed network (Fibre Channel/ iSCSI) se jodti hai.

क्यों**:** SAN servers और storage को dedicated high-speed network से जोड़ती है।

Nukte: CAN = campus buildings; PAN = personal devices; MAN = city — ye sab general-purpose networks hain.

[Source: Practice — original (textbook-based)] · Medium

Q22. ARPANET, the ancestor of today's Internet, became operational in:
आज के Internet का पूर्वज ARPANET किस वर्ष चालू हुआ?

✓ **(A) 1969**
(B) 1983
(C) 1991
(D) 1975

✓ **Answer: (A) 1969**

Why: 1969 mein ARPANET (4 nodes) chalu hua; 1983 TCP/IP; 1991 WWW.
क्यों**:** 1969 में ARPANET; 1983 TCP/IP; 1991 WWW।
Nukte: 1983 TCP/IP adopt hua tha.
[Source: Practice — original (textbook-based)] · Medium

Q23. The protocol suite used by today's Internet was developed as part of:
आज के Internet का protocol suite किस परियोजना का हिस्सा था?

(A) NSFNET
✓ **(B) ARPANET/TCP-IP project** / ARPANET/TCP-IP परियोजना
(C) Ethernet project / Ethernet परियोजना
(D) ISO project / ISO परियोजना

✓ **Answer: (B) ARPANET/TCP-IP project**

Why: TCP/IP ARPANET/DoD project se nikla; Ethernet LAN technology hai.
क्यों**:** TCP/IP ARPANET/DoD से निकला; Ethernet LAN technology है।
Nukte: ISO ne OSI model banaya.
[Source: Practice — original (textbook-based)] · Medium

Q24. The NSF backbone network that replaced ARPANET as the Internet's main backbone in the 1980s was:
1980s में ARPANET की जगह Internet का मुख्य backbone बना NSF का network था:

✓ **(A) NSFNET**
(B) MILNET
(C) BITNET
(D) USENET

✓ **Answer: (A) NSFNET**

Why: NSFNET (1986) ne supercomputer centres ko joda aur ARPANET ki jagah le li; ARPANET 1990 mein retire hua.
क्यों**:** NSFNET (1986) ने supercomputer centres जोड़े; ARPANET 1990 में हटा।
Nukte: MILNET = 1983 mein ARPANET se alag hua military hissa; BITNET/USENET = academic networks, backbone nahi.
[Source: Practice — original (textbook-based)] · Medium

Q25. In telephone networks, a dedicated end-to-end path is established before communication. This is:

Telephone network में संचार से पहले समर्पित मार्ग बनता है। यह है:

(A) Packet switching
✓ **(B) Circuit switching**
(C) Message switching
(D) Datagram switching

✓ **Answer: (B) Circuit switching**

Why: Circuit switching mein pehle dedicated circuit banta hai (telephone).

क्यों**:** Circuit switching में पहले dedicated circuit।

Nukte: Packet switching mein har packet alag route le sakta hai.

[Source: Practice — original (textbook-based)] · Medium

Q26. Which of the following statements is TRUE?

निम्न में से कौन-सा कथन सही है?

(A) The Internet and the WWW are the same thing / Internet और WWW एक ही हैं
✓ **(B) The WWW is a service that runs over the Internet** / WWW, Internet पर चलने वाली service है
(C) The Internet is a part of the WWW / Internet, WWW का हिस्सा है
(D) The WWW does not need the Internet / WWW को Internet की ज़रूरत नहीं

✓ **Answer: (B) The WWW is a service that runs over the Internet**

Why: Internet network infrastructure hai; WWW uspar chalne wali information service (web pages).

क्यों**:** Internet infrastructure; WWW उसकी service।

Nukte: Dono ko same samajhna common galti hai.

[Source: Practice — original (textbook-based)] · Hard

Q27. In ATM (Asynchronous Transfer Mode) switching, data is carried in fixed-size cells of _____ bytes:

ATM (Asynchronous Transfer Mode) switching में data _____ bytes की fixed-size cells में जाता है:

✓ **(A) 53**
(B) 64
(C) 128
(D) 1500

✓ **Answer: (A) 53**

Why: ATM cell = 53 bytes = 48-byte payload + 5-byte header — fixed size se hardware switching fast hota hai.

क्यों**:** ATM cell = 53 bytes = 48 payload + 5 header।

Nukte: 64 = Ethernet ka minimum frame; 1500 = Ethernet ka maximum payload — ye ATM ke cell size nahi.

[Source: Practice — original (textbook-based)] · Medium

Topic 2 — OSI & TCP/IP Models

Q28. In the OSI model, layers 5, 6 and 7 are mainly called:
OSI model में layers 5, 6, 7 को मुख्यतः कहा जाता है:

(A) Support layers
✓ **(B) User support layers**
(C) Network layers
(D) Physical layers

✓ **Answer: (B) User support layers**

Why: Layers 1–4 support (transport) layers; 5–7 user support layers.
क्यों**:** Layers 1–4 support; 5–7 user support।
Nukte: Support layers 1–4 hoti hain.
[Source: Practice — original (textbook-based)] · Medium

Q29. Two communicating hosts exchange data between the same layers using:
दो hosts एक ही layers के बीच data किससे बदलते हैं?

(A) Interfaces
✓ **(B) Protocols (peer-to-peer)**
(C) Cables
(D) Ports

✓ **Answer: (B) Protocols (peer-to-peer)**

Why: Same layer ka communication = peer-to-peer protocol; different layers ka apna interface.
क्यों**:** एक ही layer = peer protocol; अलग layers = interface।
Nukte: Interface upar/neeche ki layer se hota hai.
[Source: Practice — original (textbook-based)] · Medium

Q30. In the OSI model, error control as well as flow control are performed by which TWO layers?
OSI model में error control और flow control दोनों किन दो layers में होते हैं?

(A) Physical and Network / Physical और Network
✓ **(B) Data Link and Transport** / Data Link और Transport
(C) Transport and Session / Transport और Session
(D) Network and Session / Network और Session

✓ **Answer: (B) Data Link and Transport**

Why: Data link = link (node-to-node) par flow+error control; transport = end-to-end par flow+error control — dono layers ye dono kaam karti hain.
क्यों**:** Data link = link पर; transport = end-to-end — दोनों layers ये काम करती हैं।
Nukte: Network layer sirf routing/forwarding; session = dialog control.
[Source: Practice — original (textbook-based)] · Medium

Q31. Which layer adds a trailer (FCS) to the frame?
कौन-सी layer frame में trailer (FCS) जोड़ती है?

(A) Physical
✓ **(B) Data Link**
(C) Network
(D) Transport

✓ **Answer: (B) Data Link**

Why: Data link layer frame mein header + trailer (FCS) dono lagati hai.
क्यों: Data link layer header + trailer (FCS) दोनों लगाती है।
Nukte: Baaki layers sirf header.
[Source: Practice — original (textbook-based)] · Medium

Q32. A router operates at which layer of the OSI model?
Router OSI model की किस layer पर काम करता है?

(A) Data Link
✓ **(B) Network**
(C) Transport
(D) Session

✓ **Answer: (B) Network**

Why: Router = layer 3 (Network) — IP addresses par decision.
क्यों: Router = layer 3 — IP पर निर्णय।
Nukte: Switch layer 2 par.
[Source: Practice — original (textbook-based)] · Medium

Q33. Encryption and compression of data are the responsibilities of:
Data का encryption और compression किसकी ज़िम्मेदारी है?

(A) Transport layer
✓ **(B) Presentation layer**
(C) Session layer
(D) Application layer

✓ **Answer: (B) Presentation layer**

Why: Presentation layer = translate, encrypt, compress.
क्यों: Presentation = translate, encrypt, compress।
Nukte: Session dialog control karti hai.
[Source: Practice — original (textbook-based)] · Medium

Q34. Establishing, managing and terminating sessions is done by:
Sessions की स्थापना, प्रबंधन और समाप्ति किसका काम है?

✓ **(A) Session layer**
(B) Transport layer
(C) Network layer
(D) Data Link layer

✓ **Answer: (A) Session layer**

Why: Session layer = dialog control + synchronization.
क्यों: Session layer = dialog control + synchronization।
Nukte: Transport end-to-end delivery deta hai.
[Source: Practice — original (textbook-based)] · Medium

Q35. The PDU of the network layer is called:
Network layer की PDU कहलाती है:

(A) Frame
(B) Segment
✓ **(C) Packet**
(D) Bits

✓ **Answer: (C) Packet**
Why: Network layer PDU = packet/datagram; data link = frame; transport = segment.
क्यों**:** Network = packet; Data link = frame; Transport = segment।
Nukte: Frame data link ki; bits physical ki.
[Source: Practice — original (textbook-based)] · Medium

Q36. Converting the outgoing data into signals is the job of:
जाते data को signals में बदलना किसका काम है?

(A) Application layer
(B) Network layer
✓ **(C) Physical layer**
(D) Session layer

✓ **Answer: (C) Physical layer**
Why: Physical layer bits ko electrical/optical/wireless signals mein badalti hai.
क्यों**:** Physical layer bits को signals में बदलती है।
Nukte: Baaki layers data-level par kaam karti hain.
[Source: Practice — original (textbook-based)] · Medium

Q37. Dialog control and synchronization of data exchange between two applications is the responsibility of which OSI layer?
दो applications के बीच dialog control और data exchange का synchronization किस OSI layer का काम है?

✓ **(A) Session layer**
(B) Transport layer
(C) Presentation layer
(D) Application layer

✓ **Answer: (A) Session layer**
Why: Session layer dialogue manage karti hai — kaun kab bolega (dialog control), sync points lagakar recovery possible karti hai.
क्यों**:** Session layer dialogue manage करती है — कौन कब बोलेगा, sync points लगाती है।
Nukte: Transport = end-to-end delivery; presentation = format/encryption; application = user services.
[Source: Practice — original (textbook-based)] · Medium

Q38. The TCP/IP model is also known as:

TCP/IP model को और क्या कहा जाता है?

(A) OSI model
✓ **(B) DoD model**
(C) ISO model
(D) IEEE model

✓ **Answer: (B) DoD model**

Why: TCP/IP ko DoD model bhi kehte hain (US Defense of Defense se).

क्यों**:** TCP/IP = DoD model (US Defense)।

Nukte: OSI model ISO ka hai.

[Source: Practice — original (textbook-based)] · Medium

Q39. Which layer of the TCP/IP model has the protocols IP, ARP and ICMP?

TCP/IP model की किस layer में IP, ARP, ICMP protocols hote hain?

(A) Application
(B) Transport
✓ **(C) Internet**
(D) Network Access

✓ **Answer: (C) Internet**

Why: Internet layer = IP, ARP, ICMP, IGMP.

क्यों**:** Internet layer = IP, ARP, ICMP, IGMP।

Nukte: Transport mein TCP/UDP.

[Source: Practice — original (textbook-based)] · Medium

Q40. Which TCP/IP layer combines the functions of OSI's session, presentation and application layers?

TCP/IP की कौन-सी layer OSI की session, presentation और application का काम करती है?

(A) Transport
(B) Internet
✓ **(C) Application**
(D) Network Access

✓ **Answer: (C) Application**

Why: TCP/IP ki Application layer OSI ke top-3 layers ka kaam leti hai.

क्यों**:** TCP/IP की Application layer ऊपर की 3 layers का काम लेती है।

Nukte: Network Access = OSI ke bottom-2.

[Source: Practice — original (textbook-based)] · Medium

Q41. In the TCP/IP model, process-to-process (program-to-program) communication is provided by the:

TCP/IP model में process-to-process (program-to-program) संवाद किसके द्वारा होता है?

(A) Internet layer
✓ **(B) Transport layer**
(C) Network access layer
(D) Application layer

✓ **Answer: (B) Transport layer**

Why: Transport (TCP/UDP) ports ke through processes ko jodta hai; Internet layer sirf host-to-host (best-effort) delivery karti hai.

क्यों: Transport (TCP/UDP) ports से processes जोड़ता है; Internet layer host-to-host delivery।

Nukte: Yaad rakho: Network/Internet = host-to-host; Transport = process-to-process.

[Source: Practice — original (textbook-based)] · Medium

Q42. The process of adding headers (and a trailer) as data moves down the layers is called:

Layers से नीचे जाते समय headers जोड़ने की प्रक्रिया है:

(A) De-encapsulation
✓ **(B) Encapsulation**
(C) Fragmentation
(D) Modulation

✓ **Answer: (B) Encapsulation**

Why: Sender par header add = encapsulation; receiver par utarna = de-encapsulation.

क्यों: Sender पर जोड़ना = encapsulation; receiver पर उतारना = de-encapsulation।

Nukte: Fragmentation network layer ki activity hai.

[Source: Practice — original (textbook-based)] · Hard

Q43. At the receiver, the correct order of removing overheads is:

Receiver पर overheads हटाने का सही क्रम है:

✓ **(A) Physical → Data Link → Network → Transport**
(B) Transport → Network → Data Link → Physical
(C) Network → Physical → Transport → Data Link
(D) Application → Network → Physical

✓ **Answer: (A) Physical → Data Link → Network → Transport**

Why: De-encapsulation neeche se upar: bits → frame → packet → segment → data.

क्यों: De-encapsulation नीचे से ऊपर: bits → frame → packet → segment।

Nukte: Ulta kram galat hai.

[Source: Practice — original (textbook-based)] · Medium

Q44. Which combination of address and PDU is CORRECT?

Address और PDU का कौन-सा युग्म सही है?

(A) Data Link — frame — IP address
(B) Network — packet — MAC address
✓ **(C) Transport — segment — port number**
(D) Physical — frame — port number

✓ **Answer: (C) Transport — segment — port number**

Why: Transport = segment + port; Network = packet + IP; Data link = frame + MAC.

क्यों: Transport = segment + port; Network = packet + IP; Data link = frame + MAC।

Nukte: Baaki combos ulte hain.

[Source: Practice — original (textbook-based)] · Medium

Q45. On the sending host, encapsulation adds headers in which order as the message moves down the OSI stack?

Sending host पर message नीचे OSI stack में जाते समय headers किस क्रम में जुड़ते हैं?

✓ **(A) Application → Transport → Network → Data Link**
(B) Data Link → Network → Transport → Application
(C) Network → Application → Transport → Data Link
(D) Transport → Application → Data Link → Network

✓ **Answer: (A) Application → Transport → Network → Data Link**

Why: Data upar se neeche jaata hai: pehle application data, phir transport header (segment), network header (datagram), data-link header+trailer (frame). Receiver par ulta (bottom-up) order mein headers hatti hain.

क्यों: ऊपर से नीचे: application data → transport → network → data-link; receiver पर उल्टे क्रम में।

Nukte: Encapsulation = har layer apna header add karti hai; decapsulation = reverse.

[Source: Practice — original (textbook-based)] · Medium

Topic 3 — Internetworking & Addressing

Q46. The address used by the data link layer to deliver a frame across one link is called:

Data link layer द्वारा एक link पर frame पहुँचाने के लिए प्रयुक्त address है:

(A) Logical address
✓ **(B) Physical address**
(C) Port address
(D) E-mail address

✓ **Answer: (B) Physical address**

Why: MAC = physical address; hop-to-hop delivery ke liye.

क्यों: MAC = physical address; hop-to-hop के लिए।

Nukte: IP logical (end-to-end) hai.

[Source: Practice — original (textbook-based)] · Medium

Q47. How many hex digits does a 48-bit MAC address contain?
48-bit MAC address में कितने hex digits होते हैं?

(A) 6
(B) 8
✓ **(C) 12**
(D) 16

✓ **Answer: (C) 12**
Why: 48 bits ÷ 4 = 12 hex digits (jaise 00:1A:2B:3C:4D:5E).
क्यों: 48 ÷ 4 = 12 hex digits।
Nukte: 6 bytes hote hain, hex digits 12.
[Source: Practice — original (textbook-based)] · Hard

Q48. As a frame moves from hop to hop, which addresses change?
Frame के hop-to-hop जाते समय कौन-से addresses बदलते हैं?

(A) IP addresses
✓ **(B) MAC addresses**
(C) Port numbers
(D) E-mail addresses

✓ **Answer: (B) MAC addresses**
Why: Har hop par naye link ke liye source/dest MAC badalte hain; IP same.
क्यों: हर hop पर MAC बदलते हैं; IP वही।
Nukte: IP end-to-end same rehta hai.
[Source: Practice — original (textbook-based)] · Medium

Q49. MAC (hardware) addresses are assigned to NIC manufacturers and administered by:
MAC (hardware) addresses NIC manufacturers को दी जाती हैं और इनका प्रबंधन करता है:

✓ **(A) IEEE**
(B) IANA
(C) ICANN
(D) ITU

✓ **Answer: (A) IEEE**
Why: IEEE har manufacturer ko OUI (3-byte prefix) allot karta hai; baaki bytes manufacturer khud lagata hai.
क्यों: IEEE हर manufacturer को OUI (3 bytes) देता है; बाकी bytes वह खुद लगाता है।
Nukte: IANA = IP addresses/port numbers; ICANN = domain names; ITU = telecom standards.
[Source: Practice — original (textbook-based)] · Medium

Q50. Which of the following is a Class C IP address?
निम्न में से कौन-सा Class C IP address है?

(A) 10.1.1.1
(B) 128.5.5.5
✓ **(C) 192.168.1.1**
(D) 225.1.1.1

✓ **Answer: (C) 192.168.1.1**
Why: 192–223 first octet = Class C; 10.x = A; 128.x = B; 225.x = D (multicast).
क्यों: 192–223 = Class C; 10.x = A; 128.x = B; 225.x = D।
Nukte: 225 Class D multicast hai.
[Source: Practice — original (textbook-based)] · Medium

Q51. Which of the following is NOT a private IP address range?
निम्न में से कौन private IP range नहीं है?

(A) 10.x.x.x
(B) 172.16.x.x – 172.31.x.x
(C) 192.168.x.x
✓ **(D) 172.32.x.x**

✓ **Answer: (D) 172.32.x.x**
Why: Private: 10.x, 172.16–31.x, 192.168.x; 172.32 bahar hai.
क्यों**:** Private: 10.x, 172.16–31.x, 192.168.x; 172.32 बाहर।
Nukte: 172.16–31 hi private hai, 32 nahi.
[Source: Practice — original (textbook-based)] · Hard

Q52. A computer that cannot reach the DHCP server will normally get an address in the range:
जो computer DHCP server तक नहीं पहुँच पाता, उसे किस range का address मिलता है?

(A) 127.x.x.x
✓ **(B) 169.254.x.x**
(C) 224.x.x.x
(D) 0.0.0.0

✓ **Answer: (B) 169.254.x.x**
Why: APIPA (169.254.x.x) automatically assign hota hai jab DHCP fail ho.
क्यों**:** APIPA (169.254.x.x) — DHCP fail होने पर।
Nukte: 127 loopback hai.
[Source: Practice — original (textbook-based)] · Medium

Q53. An octet in an IPv4 address can have a value between:
IPv4 address में एक octet का मान हो सकता है:

✓ **(A) 0 to 255** / 0 से 255
(B) 0 to 128 / 0 से 128
(C) 1 to 256 / 1 से 256
(D) 0 to 511 / 0 से 511

✓ **Answer: (A) 0 to 255**
Why: 8 bits = 2^8 = 256 values = 0–255.
क्यों**:** 8 bits = 2^8 = 256 मान = 0–255।
Nukte: 256 nahi 255 tak.
[Source: Practice — original (textbook-based)] · Medium

Q54. How many usable host addresses does a Class A network (default mask) have?
Class A network (default mask) में कितने usable host addresses होते हैं?

(A) 2^{24}
✓ **(B)** $2^{24} - 2$
(C) $2^{16} - 2$
(D) $2^8 - 2$

✓ **Answer: (B)** $2^{24} - 2$
Why: Class A: 24 host bits → $2^{24} - 2$ usable.
क्यों**:** Class A: 24 host bits → $2^{24} - 2$।
Nukte: 2^{24} mein network aur broadcast bhi gine hain.
[Source: Practice — original (textbook-based)] · Hard

Q55. The IPv4 address block 100.64.0.0/10 is reserved by IANA for:
IPv4 address block 100.64.0.0/10 IANA द्वारा किसके लिए आरक्षित है?

✓ **(A) Carrier-grade NAT (shared address space)**
(B) Loopback testing
(C) Multicasting
(D) Experimental use

✓ **Answer: (A) Carrier-grade NAT (shared address space)**
Why: 100.64.0.0/10 = CGNAT/ISP ke andar private addressing (RFC 6598) — jab ISP ke paas public IPs kam hon. 127=loopback, 224=multicast, 240/4=experimental.
क्यों: 100.64.0.0/10 = CGNAT (RFC 6598) — ISP आंतरिक addressing; 127=loopback, 224=multicast।
Nukte: Ye IANA special-purpose registry ka block hai — worldwide exams mein pucha jaata hai.
[Source: Practice — original (textbook-based)] · Medium

Q56. With a /26 subnet mask, how many usable host addresses are there per subnet?
/26 subnet mask पर हर subnet में कितने usable hosts होंगे?

(A) 30
✓ **(B) 62**
(C) 64
(D) 126

✓ **Answer: (B) 62**
Why: /26 = 6 host bits → $2^6 - 2 = 62$.
क्यों: /26 = 6 host bits → $2^6 - 2 = 62$।
Nukte: 64 total hai, usable 62.
[Source: Practice — original (textbook-based)] · Hard

Q57. The subnet mask 255.255.255.224 divides a /24 network into subnets of size:
Mask 255.255.255.224, /24 network को कितने size के subnets में बाँटता है?

✓ **(A) 32 addresses**
(B) 64 addresses
(C) 16 addresses
(D) 8 addresses

✓ **Answer: (A) 32 addresses**
Why: 224 = 11100000 → 5 host bits → block size $2^5 = 32$.
क्यों: 224 → 5 host bits → block 32।
Nukte: 64 block /26 ka hota hai.
[Source: Practice — original (textbook-based)] · Hard

Q58. In NAT overload (PAT), many private hosts share:
NAT overload (PAT) में कई private hosts साझा करते हैं:

✓ **(A) One public IP with different port numbers** / एक public IP, अलग-अलग ports के साथ
(B) One private IP / एक private IP
(C) Different public IPs / अलग-अलग public IPs
(D) No IP at all / कोई IP नहीं

✓ **Answer: (A) One public IP with different port numbers**
Why: PAT = ek public IP + alag ports se sab hosts.
क्यों: PAT = एक public IP + अलग ports।
Nukte: Static NAT mein 1-1 mapping hoti hai.
[Source: Practice — original (textbook-based)] · Medium

Q59. Route aggregation (summarization) is used to:
Route aggregation का प्रयोग होता है:

(A) Increase security / Security बढ़ाने के लिए
✓ **(B) Reduce the size of routing tables** / Routing tables छोटी करने के लिए
(C) Increase speed of cables / Cables की speed बढ़ाने के लिए
(D) Encrypt the packets / Packets encrypt करने के लिए

✓ **Answer: (B) Reduce the size of routing tables**
Why: CIDR summarization se kai routes ek entry mein aa jaati hain.
क्यों**:** CIDR summarization से कई routes एक entry में।
Nukte: Security firewall ka kaam.
[Source: Practice — original (textbook-based)] · Medium

Q60. A network 192.168.10.0/24 is divided using /27. How many subnets are created?
192.168.10.0/24 को /27 से बाँटा गया। कितने subnets बनेंगे?

(A) 4
✓ **(B) 8**
(C) 16
(D) 32

✓ **Answer: (B) 8**
Why: Borrowed bits = 27 − 24 = 3 → 2^3 = 8 subnets.
क्यों**:** Borrowed = 3 → 2^3 = 8।
Nukte: 32 subnets tab jab /29 jaisa ho.
[Source: Practice — original (textbook-based)] · Hard

Q61. In VLSM, different subnets of the same network use:
VLSM में एक ही network के अलग-अलग subnets प्रयोग करते हैं:

(A) The same mask everywhere / सब जगह एक ही mask
✓ **(B) Masks of different lengths** / अलग-अलग लंबाई के masks
(C) No mask / कोई mask नहीं
(D) Only /24 / केवल /24

✓ **Answer: (B) Masks of different lengths**
Why: VLSM = Variable Length Subnet Mask — zaroorat ke hisaab se alag size.
क्यों**:** VLSM = अलग लंबाई के masks।
Nukte: Same mask = FLSM (fixed).
[Source: Practice — original (textbook-based)] · Hard

Q62. The usable host range of the CIDR block 172.16.45.64/26 is:
CIDR block 172.16.45.64/26 की usable host range है:

✓ **(A) 172.16.45.65 – 172.16.45.126**
(B) 172.16.45.65 – 172.16.45.127
(C) 172.16.45.64 – 172.16.45.127
(D) 172.16.45.1 – 172.16.45.254

✓ **Answer: (A) 172.16.45.65 – 172.16.45.126**
Why: /26 = 26 network bits → block size 64. 64 se 127 tak ka block: network=64, broadcast=127, beech ke 62 addresses usable (65–126).
क्यों**:** /26 → block size 64; network=64, broadcast=127; usable = 65–126 (62 addresses)।
Nukte: Broadcast address (127) aur network address (64) hosts ko nahi milte.
[Source: Practice — original (textbook-based)] · Hard

Q63. For point-to-point links between routers, a /30 prefix is preferred because it provides exactly _____ usable host addresses:

Routers के बीच point-to-point links के लिए /30 prefix क्यों पसंद किया जाता है? यह ठीक _____ usable addresses देता है:

✓ **(A) 2**
(B) 4
(C) 6
(D) 14

✓ **Answer: (A) 2**

Why: /30 = 4 addresses: network + broadcast + 2 hosts — P2P link par sirf 2 interfaces hote hain, isliye ideal.

क्यों**:** /30 = 4 addresses: network + broadcast + 2 hosts — P2P link के लिए ideal।

Nukte: 2^2=4 addresses total; 2 reserved; 2 usable.

[Source: Practice — original (textbook-based)] · Medium

Q64. An organization has the block 192.168.10.0/24 and needs subnets for 50, 25 and 10 hosts. Which set of prefix lengths wastes the LEAST address space?

एक संस्था के पास block 192.168.10.0/24 है और उसे 50, 25 और 10 hosts के लिए subnets चाहिए। कौन-सी prefix lengths की जोड़ी सबसे कम address space व्यर्थ करेगी?

✓ **(A) /26, /27 and /28** / /26, /27 और /28
(B) /26, /26 and /26 / /26, /26 और /26
(C) /27, /27 and /27 / /27, /27 और /27
(D) /25, /26 and /27 / /25, /26 और /27

✓ **Answer: (A) /26, /27 and /28**

Why: 50 hosts → /26 (62 usable), 25 → /27 (30), 10 → /28 (14). Har subnet ko sabse chhota fitting power-of-2 block do. Pattern: RPSC Programmer/ACP/DD jaise Rajasthan technical exams mein VLSM sizing aksar aata hai.

क्यों**:** 50 → /26 (62), 25 → /27 (30), 10 → /28 (14)। Pattern: RPSC Programmer/ACP/DD में VLSM sizing अक्सर आता है।

Nukte: VLSM rule: hosts + 2 ≤ block size, aur block size = 2^(32−prefix).

[Source: Practice — original (RPSC Programmer/ACP/DD exam-pattern, textbook-based)] · Hard

Q65. The Cisco IOS command 'ip classless' makes a router:

Cisco IOS का आदेश 'ip classless' router को क्या बनाता है?

✓ **(A) Forward packets to the default route even when the destination subnet is not in the routing table** / तब भी default route पर packets भेजे जब destination subnet routing table में न हो
(B) Drop all packets for classful networks / Classful networks के सभी packets गिराए
(C) Advertise only classful routes / सिर्फ classful routes advertise करे
(D) Disable subnetting on all interfaces / सभी interfaces पर subnetting बंद करे

✓ **Answer: (A) Forward packets to the default route even when the destination subnet is not in the routing table**

Why: Classless forwarding: routing table mein koi specific subnet na mile to packet default route par jaata hai (supernetting/CDR ke saath zaroori). Pattern: RPSC Programmer/ACP/DD + CCNA.

क्यों**:** Classless forwarding: विशेष subnet न मिले तो packet default route पर जाता है। Pattern: RPSC Programmer/ACP/DD।

Nukte: 'Classless' ka matlab subnet masks ko ignore karke longest-match karna.

[Source: Practice — original (RPSC Programmer/ACP/DD exam-pattern, textbook-based)] · Hard

Q66. Which technique carries an IPv6 packet inside an IPv4 packet?
कौन-सी technique IPv6 packet को IPv4 के अंदर ले जाती है?

(A) Dual stack
✓ **(B) Tunneling**
(C) Fragmentation
(D) NAT

✓ **Answer: (B) Tunneling**

Why: Tunneling = IPv6 ko IPv4 header ke andar bhejna (transition).
क्यों: Tunneling = IPv6 को IPv4 के अंदर।
Nukte: Dual stack dono IP stack chalata hai.
[Source: Practice — original (textbook-based)] · Hard

Q67. Which statement about the IPv6 header is TRUE?
IPv6 header के बारे में कौन-सा कथन सही है?

✓ **(A) It has a fixed size of 40 bytes** / इसका size 40 bytes fixed है
(B) It is variable from 20 to 60 bytes / यह 20 से 60 bytes तक बदलता है
(C) It contains a header checksum / इसमें header checksum होता है
(D) It has a broadcast flag / इसमें broadcast flag होता है

✓ **Answer: (A) It has a fixed size of 40 bytes**

Why: IPv6 header fixed 40 bytes; checksum nahi; broadcast flag nahi.
क्यों: IPv6 header 40 bytes fixed; checksum नहीं।
Nukte: 20–60 bytes IPv4 ka hai.
[Source: Practice — original (textbook-based)] · Hard

Q68. In IPv6, a host can generate its own address automatically without any DHCP server using:
IPv6 में host बिना DHCP server के अपना address स्वयं बना सकता है — किसके द्वारा?

✓ **(A) SLAAC (Stateless Address Autoconfiguration)**
(B) ARP
(C) BOOTP
(D) NAT-PT

✓ **Answer: (A) SLAAC (Stateless Address Autoconfiguration)**

Why: SLAAC: router ke RA (Router Advertisement) se prefix milta hai aur host apna Interface ID (EUI-64/ random) khud jod leta hai — stateless.
क्यों: SLAAC: router के RA से prefix मिलता है, host अपना Interface ID जोड़ लेता है।
Nukte: ARP = IPv4 MAC resolution; BOOTP = DHCP ka purana version; NAT-PT = IPv4-IPv6 translation.
[Source: Practice — original (textbook-based)] · Medium

Q69. The prefix used for IPv6 link-local addresses (usable only on the local link) is:
IPv6 link-local addresses (सिर्फ उसी link पर काम करने वाले) का prefix होता है:

✓ **(A) FE80::/10**
(B) 2000::/3
(C) FF00::/8
(D) ::1/128

✓ **Answer: (A) FE80::/10**

Why: FE80::/10 = link-local (har interface par automatically); 2000::/3 = global unicast; FF00::/8 = multicast; ::1 = loopback.

क्यों**:** FE80::/10 = link-local; 2000::/3 = global unicast; FF00::/8 = multicast; ::1 = loopback।

Nukte: Link-local = bina router ke same-link communication (neighbour discovery).

[Source: Practice — original (textbook-based)] · Medium

Q70. What is the range of well-known port numbers?
Well-known port numbers की range है:

✓ **(A) 0 – 1023**
(B) 1024 – 49151
(C) 49152 – 65535
(D) 0 – 65535

✓ **Answer: (A) 0 – 1023**

Why: Well-known 0–1023; registered 1024–49151; ephemeral 49152–65535.

क्यों**:** Well-known 0–1023; registered; ephemeral।

Nukte: 0–65535 poori range hai.

[Source: Practice — original (textbook-based)] · Medium

Q71. Which DNS record type is used to find the domain name from an IP address?
IP से domain name जानने के लिए कौन-सा DNS record प्रयोग होता है?

(A) A
(B) MX
✓ **(C) PTR**
(D) CNAME

✓ **Answer: (C) PTR**

Why: PTR = reverse lookup (IP → naam); A = naam → IPv4.

क्यों**:** PTR = reverse lookup; A = नाम → IPv4।

Nukte: MX mail ke liye; CNAME alias.

[Source: Practice — original (textbook-based)] · Medium

Q72. A socket address is the combination of:
Socket address किसका संयोजन है?

(A) MAC address and IP address / MAC और IP address
✓ **(B) IP address and port number** / IP address और port number
(C) MAC address and port number / MAC और port number
(D) URL and e-mail address / URL और e-mail address

✓ **Answer: (B) IP address and port number**

Why: Socket = IP + port — process ko identify karta hai.

क्यों**:** Socket = IP + port।

Nukte: MAC data link level par.

[Source: Practice — original (textbook-based)] · Medium

Q73. DNS zone transfers between primary and secondary name servers use:
Primary और secondary name servers के बीच DNS zone transfer प्रयोग करती है:

✓ **(A) TCP port 53**
(B) UDP port 53
(C) UDP port 69
(D) TCP port 25

✓ **Answer: (A) TCP port 53**

Why: Normal DNS query = UDP 53 (chhota request/response); zone transfer = pura zone ka bulk data — reliable TCP 53 chahiye.

क्यों**:** सामान्य query = UDP 53; zone transfer = bulk data — इसलिए TCP 53।

Nukte: UDP 53 = queries; UDP 69 = TFTP; TCP 25 = SMTP.

[Source: Practice — original (textbook-based)] · Medium

Topic 4 — Connecting LANs, Switching & Routing

Q74. All ports of a hub belong to how many collision domains?
Hub के सभी ports कितने collision domains में आते हैं?

✓ **(A) One** / एक
(B) Equal to number of ports / Ports की संख्या के बराबर
(C) Two / दो
(D) Zero / शून्य

✓ **Answer: (A) One**

Why: Hub ek shared medium hai — sab ports ek hi collision domain.

क्यों**:** Hub में सब ports एक collision domain।

Nukte: Switch mein har port alag.

[Source: Practice — original (textbook-based)] · Hard

Q75. A device that regenerates a weak signal bit-by-bit to its original strength is a:
कमज़ोर signal को bit-by-bit पुनः मूल शक्ति तक लाने वाला उपकरण है:

✓ **(A) Repeater**
(B) Amplifier
(C) Bridge
(D) Router

✓ **Answer: (A) Repeater**

Why: Repeater signal copy karke regenerate karta hai — amplify nahi.

क्यों**:** Repeater regenerate करता है — amplify नहीं।

Nukte: Amplifier shor ke saath badhata hai.

[Source: Practice — original (textbook-based)] · Medium

Q76. A 24-port hub creates how many collision domains?
24-port hub कितने collision domains बनाता है?

✓ **(A) 1**
(B) 12
(C) 24
(D) 48

✓ **Answer: (A) 1**

Why: Hub ek shared segment hai — saare ports ek hi collision domain mein hain (switch hota to har port apna collision domain).

क्यों**:** Hub एक shared segment है — सारे ports एक ही collision domain में (switch में हर port अलग)।

Nukte: Hub = layer-1 dumb device, signal ko sab ports par repeat karta hai.

[Source: Practice — original (textbook-based)] · Hard

Q77. A switch learns the port for a MAC address by:
Switch किसी MAC address का port कैसे सीखता है?

✓ **(A) Reading the source MAC of incoming frames** / आने वाले frames के source MAC से
(B) Asking the router / Router से पूछकर
(C) Reading the IP address / IP address पढ़कर
(D) Broadcasting every frame / हर frame broadcast करके

✓ **Answer: (A) Reading the source MAC of incoming frames**

Why: Switch incoming frame ka source MAC dekh kar CAM table bharta hai.

क्यों**:** Switch source MAC देखकर CAM table भरता है।

Nukte: Destination MAC se forwarding hoti hai, learning source se.

[Source: Practice — original (textbook-based)] · Medium

Q78. VLANs on a switch are used to create:
Switch पर VLAN बनाए जाते हैं:

(A) More collision domains / अधिक collision domains
✓ **(B) Separate broadcast domains** / अलग-अलग broadcast domains
(C) More repeaters / अधिक repeaters
(D) Faster cables / तेज़ cables

✓ **Answer: (B) Separate broadcast domains**

Why: VLAN = ek switch par alag broadcast domains.

क्यों**:** VLAN = अलग broadcast domains।

Nukte: Collision domain to switch pehle hi har port par deta hai.

[Source: Practice — original (textbook-based)] · Medium

Q79. Compared to a hub, a switch:
Hub की तुलना में switch:

(A) Sends every frame to all ports / हर frame सब ports पर भेजता है
✓ **(B) Sends frames only to the destination port** / Frame सिर्फ़ destination port पर भेजता है
(C) Works only at layer 3 / सिर्फ़ layer 3 पर काम करता है
(D) Does not need MAC addresses / MAC addresses की ज़रूरत नहीं

✓ **Answer: (B) Sends frames only to the destination port**

Why: Switch MAC table se frame ko sahi port par bhejta hai.

क्यों**:** Switch MAC table से सही port पर भेजता है।

Nukte: Hub sabko bhejta hai.

[Source: Practice — original (textbook-based)] · Hard

Q80. Connecting each switch port to a single host so that every port becomes its own collision domain is called:

Switch के हर port पर एक host लगाकर उसे अलग collision domain बनाना कहलाता है:

✓ **(A) Microsegmentation**
(B) Trunking
(C) VLAN hopping
(D) Bridging loop

✓ **Answer: (A) Microsegmentation**

Why: Microsegmentation se collisions khatam — full-duplex possible; trunking = multiple VLANs ka traffic ek link par.

क्यों**:** Microsegmentation से collisions खत्म; trunking = कई VLANs का traffic एक link पर।

Nukte: Collision domain ka size chhota karna hi microsegmentation hai.

[Source: Practice — original (textbook-based)] · Medium

Q81. On a managed switch, which feature restricts a port to specific MAC addresses so that unauthorized hosts cannot connect?

Managed switch पर कौन-सी सुविधा एक port को तय MAC addresses तक सीमित करती है ताकि अनधिकित host न जुड़ सकें?

✓ **(A) Port Security**
(B) VTP pruning
(C) Spanning Tree
(D) Port mirroring

✓ **Answer: (A) Port Security**

Why: Port Security: ek port par allowed MAC addresses (sticky/static) fix karo; adhik aane par port err-disable ho jaata hai. Pattern: SSC CGL + CCNA/Network+.

क्यों**:** Port Security: port पर allowed MAC तय; उल्लंघन पर port बंद। Pattern: SSC CGL + CCNA।

Nukte: VTP = VLAN config distribute; STP = loops roke; mirroring = traffic copy.

[Source: Practice — original (SSC CGL/IBPS exam-pattern, textbook-based)] · Medium

Q82. When a router forwards a packet, which header field does NOT change?

Router के packet forward करने पर कौन-सा header field नहीं बदलता?

(A) TTL
(B) Header checksum
✓ **(C) Source IP address**
(D) Both TTL and checksum / TTL और checksum दोनों

✓ **Answer: (C) Source IP address**

Why: TTL ghat-ta hai, checksum naya hota hai; source IP same.

क्यों**:** TTL घटता है, checksum नया; source IP same।

Nukte: TTL/checksum dono badalte hain.

[Source: Practice — original (textbook-based)] · Hard

Q83. A brouter is a device that can act as:

Brouter वह उपकरण है जो बन सकता है:

(A) Hub + repeater
✓ **(B) Bridge + router**
(C) Switch + modem
(D) Firewall + gateway

✓ **Answer: (B) Bridge + router**

Why: Brouter = bridge + router dono ki functionality.

क्यों**:** Brouter = bridge + router।

Nukte: Baaki combos asli devices nahi.

[Source: Practice — original (textbook-based)] · Medium

Q84. A brouter forwards non-routable protocols (e.g., NetBEUI) at OSI layer _____ and routable protocols (e.g., IP) at layer 3:

Brouter non-routable protocols (जैसे NetBEUI) को OSI layer _____ पर और routable protocols (IP) को layer 3 पर भेजता है:

(A) 1
✓ **(B) 2**
(C) 3
(D) 4

✓ **Answer: (B) 2**

Why: Brouter = bridge + router: jin protocols par routing possible nahi unke liye bridge (layer 2) ki tarah; routable ke liye router (layer 3) ki tarah.

क्यों**:** Brouter = bridge + router: गैर-routable के लिए bridge (layer 2), routable के लिए router (layer 3)।

Nukte: Bridge ka kaam = MAC addresses se frame forward karna.

[Source: Practice — original (textbook-based)] · Medium

Q85. A firewall that keeps track of the state of live connections is called:

वह firewall जो चालू connections की state याद रखता है, कहलाता है:

(A) Packet-filter firewall
✓ **(B) Stateful firewall**
(C) Repeater
(D) Modem

✓ **Answer: (B) Stateful firewall**

Why: Stateful firewall connection ka state table rakhta hai.

क्यों**:** Stateful firewall state table रखता है।

Nukte: Packet filter sirf headers dekhta hai.

[Source: Practice — original (textbook-based)] · Medium

Q86. The area where public-facing servers (web, mail) are placed for security is called:
जहाँ public servers (web, mail) सुरक्षा हेतु रखे जाते हैं, वह क्षेत्र है:

(A) LAN
(B) VLAN
✓ **(C) DMZ**
(D) SAN

✓ **Answer: (C) DMZ**
Why: DMZ = Demilitarized Zone — firewall ke beech ka public area.
क्यों**:** DMZ = Demilitarized Zone।
Nukte: VLAN switch ki virtual LAN hai.
[Source: Practice — original (textbook-based)] · Medium

Q87. Which NAT type uses a single public IP for many internal hosts?
कौन-सी NAT कई internal hosts के लिए एक ही public IP प्रयोग करती है?

(A) Static NAT
(B) Dynamic NAT
✓ **(C) Overload / PAT**
(D) No NAT

✓ **Answer: (C) Overload / PAT**
Why: PAT ports se differentiate karta hai.
क्यों**:** PAT ports से भेद करता है।
Nukte: Static mein fixed 1-1.
[Source: Practice — original (textbook-based)] · Medium

Q88. Port Address Translation (PAT), which maps many private IP addresses to one public IP using port numbers, is also called:
Port Address Translation (PAT) — जो port numbers से कई private IPs को एक public IP पर map करती है — कहलाती है भी:

✓ **(A) NAT overload**
(B) Static NAT
(C) Dynamic NAT
(D) Proxy NAT

✓ **Answer: (A) NAT overload**
Why: Cisco terminology: 'ip nat inside source list … overload' = PAT = NAT overload — ek public IP se 65536 tak connections.
क्यों**:** Cisco में '... overload' = PAT = NAT overload — एक public IP से हजारों connections।
Nukte: Static NAT = 1:1 fixed mapping; dynamic NAT = pool se temporary 1:1.
[Source: Practice — original (textbook-based)] · Medium

Topic 5 — Telephony, TV, Cable & Wireless

Q89. According to Nyquist, the maximum bit rate of a noiseless channel with bandwidth 3000 Hz and 4 signal levels is:

Nyquist के अनुसार 3000 Hz bandwidth, 4 signal levels वाले noiseless channel की अधिकतम bit rate है:

(A) 6000 bps
✓ **(B) 12000 bps**
(C) 24000 bps
(D) 3000 bps

✓ **Answer: (B) 12000 bps**

Why: $2 \times B \times \log_2 L = 2 \times 3000 \times 2 = 12000$ bps.

क्यों: $2 \times 3000 \times 2 = 12000$ bps।

Nukte: 6000 $\log_2 L$ bhool kar; 24000 double-count.

[Source: Practice — original (textbook-based)] · Hard

Q90. A signal-to-noise ratio of 1000 corresponds to how many dB?

Signal-to-noise ratio 1000 कितने dB है?

(A) 3 dB
(B) 10 dB
✓ **(C) 30 dB**
(D) 100 dB

✓ **Answer: (C) 30 dB**

Why: SNR(dB) = $10 \log_{10}(1000) = 30$ dB.

क्यों: $10 \log_{10}(1000) = 30$ dB।

Nukte: 10 dB = ratio 10.

[Source: Practice — original (textbook-based)] · Hard

Q91. Which encoding has a transition in the middle of every bit and is self-synchronizing?

कौन-सी encoding में हर bit के बीच transition होता है और वह self-synchronizing है?

(A) NRZ-L
✓ **(B) Manchester**
(C) Unipolar
(D) Analog

✓ **Answer: (B) Manchester**

Why: Manchester: bit ke beech transition — clock recovery aasan.

क्यों: Manchester: bit के बीच transition।

Nukte: NRZ level constant rakhta hai.

[Source: Practice — original (textbook-based)] · Medium

Q92. Attenuation of a signal means:
Signal का attenuation अर्थ है:

✓ **(A) Its strength decreases with distance** / दूरी के साथ शक्ति घटना
(B) Its frequency increases / आवृत्ति बढ़ना
(C) It becomes digital / वह digital होना
(D) It gets an error / उसमें त्रुटि आना

✓ **Answer: (A) Its strength decreases with distance**
Why: Attenuation = signal weak hona distance ke saath; repeater se restore.
क्यों**:** Attenuation = दूरी से कमज़ोरी।
Nukte: Frequency ka badalna distortion kehlata hai.
[Source: Practice — original (textbook-based)] · Medium

Q93. Crosstalk is a type of:
Crosstalk किसका प्रकार है?

✓ **(A) Noise**
(B) Encoding
(C) Protocol
(D) Topology

✓ **Answer: (A) Noise**
Why: Crosstalk = ek wire ka signal doosre mein interfere karna — ek noise.
क्यों**:** Crosstalk = एक wire का signal दूसरे में — एक noise।
Nukte: Encoding data ka format hai.
[Source: Practice — original (textbook-based)] · Medium

Q94. According to Shannon's capacity theorem, a 1 MHz channel with SNR = 30 dB has a capacity of approximately:
Shannon के सूत्र से 1 MHz channel जिसका SNR = 30 dB हो, की capacity लगभग होगी:

✓ **(A) 10 Mbps**
(B) 20 Mbps
(C) 30 Mbps
(D) 1 Mbps

✓ **Answer: (A) 10 Mbps**
Why: SNR(dB) = 30 → SNR = 1000. C = B × log2(1+SNR) = 10^6 × log2(1001) ≈ 10^6 × 9.97 ≈ 10 Mbps.
क्यों**:** SNR(dB)=30 → SNR=1000; C = 10^6 × log2(1001) ≈ 10 Mbps।
Nukte: Shannon mein SNR linear value lagti hai, dB nahi — pehle convert karo.
[Source: Practice — original (textbook-based)] · Hard

Q95. In NRZ-I encoding, a binary 1 is represented by:

NRZ-I encoding में binary 1 को दर्शाया जाता है:

✓ **(A) A transition at the beginning of the bit interval** / Bit interval की शुरुआत में transition

(B) No transition at all / कोई transition नहीं

(C) A transition in the middle of the bit / Bit के बीच में transition

(D) A positive voltage / धनात्मक voltage

✓ **Answer: (A) A transition at the beginning of the bit interval**

Why: NRZ-I (Invert): 1 = signal invert (transition) at start of bit; 0 = no transition. Consecutive 1s ke liye clock sync rehti hai.

क्यों: NRZ-I: 1 = bit की शुरुआत में invert/transition; 0 = बिना transition।

Nukte: Middle transition = Manchester (har bit mein); NRZ-I mein middle nahi.

[Source: Practice — original (textbook-based)] · Medium

Q96. A noiseless channel has a bandwidth of 4 kHz and uses 16 signal levels. Per Nyquist's theorem, the maximum bit rate is:

एक noiseless channel की bandwidth 4 kHz है और वह 16 signal levels प्रयोग करती है। Nyquist के अनुसार अधिकतम bit rate होगी:

✓ **(A) 32 kbps**

(B) 16 kbps

(C) 64 kbps

(D) 8 kbps

✓ **Answer: (A) 32 kbps**

Why: Nyquist: C = 2 × B × log2(L) = 2 × 4000 × log2(16) = 2 × 4000 × 4 = 32 kbps. Pattern: RPSC Programmer/DD + GATE-style signal numerics.

क्यों: Nyquist: C = 2 × 4000 × log2(16) = 32 kbps। Pattern: RPSC Programmer/DD + GATE-style।

Nukte: Shannon (noise wala) = B log2(1+SNR); Nyquist (noiseless) = 2B log2(L) — mix mat karna.

[Source: Practice — original (RPSC Programmer/ACP/DD exam-pattern, textbook-based)] · Hard

Q97. An analog voice signal is sampled 8000 times per second with 8 bits per sample. The resulting digital bit rate is:

एक analog voice signal को प्रति सेकंड 8000 बार sample किया जाता है, हर sample 8 bits का। बनी digital bit rate होगी:

✓ **(A) 64 kbps**

(B) 8 kbps

(C) 56 kbps

(D) 128 kbps

✓ **Answer: (A) 64 kbps**

Why: 8000 samples/s × 8 bits = 64,000 bps = 64 kbps — yahi classic PCM/DS0 voice channel hai (telephony standard). Pattern: SSC CGL/IBPS + technical exams.

क्यों: 8000 × 8 = 64 kbps — यही PCM/DS0 voice channel है। Pattern: SSC CGL/IBPS।

Nukte: Nyquist: voice ke liye 8 kHz sampling enough (4 kHz bandwidth × 2).

[Source: Practice — original (SSC CGL/IBPS exam-pattern, textbook-based)] · Hard

Q98. Which fiber type is used for long-distance communication?
लंबी दूरी के संचार के लिए कौन-सी fiber प्रयोग होती है?

(A) Multimode fiber
✓ **(B) Single-mode fiber**
(C) Coaxial cable
(D) UTP

✓ **Answer: (B) Single-mode fiber**
Why: Single-mode: ek light ray, kam dispersion — lambi doori.
क्यों**:** Single-mode: एक ray — लंबी दूरी।
Nukte: Multimode chhoti doori ke liye.
[Source: Practice — original (textbook-based)] · Medium

Q99. UTP cables are normally terminated with which connector?
UTP cables सामान्यतः किस connector से जुड़ती हैं?

(A) BNC
✓ **(B) RJ-45**
(C) SC
(D) USB

✓ **Answer: (B) RJ-45**
Why: UTP → RJ-45; coaxial → BNC; fiber → SC/LC.
क्यों**:** UTP → RJ-45; coaxial → BNC; fiber → SC/LC।
Nukte: BNC coaxial ka.
[Source: Practice — original (textbook-based)] · Medium

Q100. Cable TV networks commonly use which cable impedance?
Cable TV networks सामान्यतः किस impedance की cable प्रयोग करते हैं?

(A) 50 ohm
✓ **(B) 75 ohm**
(C) 100 ohm
(D) 600 ohm

✓ **Answer: (B) 75 ohm**
Why: 75 ohm coaxial cable TV ke liye; 50 ohm Ethernet ke liye.
क्यों**:** 75 Ω cable TV; 50 Ω Ethernet।
Nukte: 100 ohm UTP ka nominal.
[Source: Practice — original (textbook-based)] · Medium

Q101. For an office LAN needing low cost and easy installation up to 100 m, the best choice is:
100 m तक, कम लागत और आसान installation वाले office LAN के लिए सबसे अच्छा विकल्प:

(A) Fiber optic
✓ **(B) UTP**
(C) Coaxial
(D) Satellite

✓ **Answer: (B) UTP**
Why: UTP sasta, aasan, 100 m tak perfect.
क्यों**:** UTP सस्ता, 100 m तक।
Nukte: Fiber mehenga; satellite bahut door.
[Source: Practice — original (textbook-based)] · Medium

Q102. Which twisted-pair cable standard supports 10 Gbps Ethernet over the full 100 m channel?

कौन-सी twisted-pair cable standard पूरे 100 m channel पर 10 Gbps Ethernet चलाती है?

(A) Cat5e
(B) Cat6
✓ **(C) Cat6a**
(D) Cat3

✓ **Answer: (C) Cat6a**

Why: Cat6a = 10 Gbps @ 100 m (augmented, alien-crosstalk controlled); Cat6 = 10 Gbps sirf ~55 m tak.

क्यों: Cat6a = 10 Gbps @ 100 m; Cat6 = 10 Gbps सिर्फ ~55 m तक।

Nukte: Cat5e = 1 Gbps; Cat3 = 10 Mbps (purani voice-grade).

[Source: Practice — original (textbook-based)] · Hard

Q103. Microwave communication requires:

Microwave communication के लिए आवश्यक है:

✓ **(A) Line of sight**
(B) Wires / तार
(C) Fiber cable
(D) Underground cable / भूमिगत cable

✓ **Answer: (A) Line of sight**

Why: Microwave towers ke beech direct line-of-sight chahiye.

क्यों: Microwave को direct line-of-sight चाहिए।

Nukte: Wires/fiber guided media hain.

[Source: Practice — original (textbook-based)] · Medium

Q104. Bluetooth is designed for which type of network?

Bluetooth किस प्रकार के network के लिए बना है?

(A) WAN
✓ **(B) PAN**
(C) SAN
(D) CAN

✓ **Answer: (B) PAN**

Why: Bluetooth = PAN (~10 m, 2.4 GHz).

क्यों: Bluetooth = PAN (~10 m)।

Nukte: WAN door ka; SAN storage.

[Source: Practice — original (textbook-based)] · Medium

Q105. A geostationary satellite orbits approximately at a height of:

Geostationary satellite लगभग किस ऊँचाई पर चक्कर लगाता है?

✓ **(A) 36,000 km**
(B) 500 km
(C) 1,000 km
(D) 100 km

✓ **Answer: (A) 36,000 km**

Why: Geostationary orbit ≈ 36,000 km (Earth ke saath ghoomta hai).

क्यों: Geostationary ≈ 36,000 km।

Nukte: 500 km low-earth orbit.

[Source: Practice — original (textbook-based)] · Hard

Q106. The IEEE 802.11n (Wi-Fi 4) standard greatly increased wireless throughput by using multiple antennas for parallel data streams, a technology called:

IEEE 802.11n (Wi-Fi 4) ने throughput बढ़ाने के लिए कई antennas से समांतर data streams भेजीं — यह तकनीक है:

✓ **(A) MIMO**
(B) OFDM
(C) DSSS
(D) FHSS

✓ **Answer: (A) MIMO**

Why: MIMO (Multiple-Input Multiple-Output) = multiple spatial streams; 802.11n se 600 Mbps tak theoretical rate; OFDM pehle 802.11a/g mein tha.

क्यों**:** MIMO = कई spatial streams; 802.11n से सैद्धांतिक 600 Mbps; OFDM 802.11a/g में था।

Nukte: OFDM = subcarriers par modulation (802.11a/g); DSSS/FHSS = purani spread spectrum techniques (802.11b).

[Source: Practice — original (textbook-based)] · Medium

Q107. The name of a wireless network (Wi-Fi) that devices see while connecting is called:

वह नाम जो devices को Wi-Fi जोड़ते समय दिखता है, कहलाता है:

✓ **(A) SSID**
(B) BSSID
(C) MAC ID
(D) IPX

✓ **Answer: (A) SSID**

Why: SSID = Service Set Identifier — wireless network ka naam (jo access point broadcast karta hai). Pattern: SSC MTS/CHSL — computer awareness level.

क्यों**:** SSID = Service Set Identifier — wireless network का नाम। Pattern: SSC MTS/CHSL।

Nukte: BSSID = access point ka MAC address; SSID = network ka naam.

[Source: Practice — original (SSC MTS/CHSL exam-pattern, textbook-based)] · Easy

Topic 6 — Transport, Routing & Application

Q108. The minimum Hamming distance required to detect 2 bit errors is:

2 bit errors detect करने के लिए न्यूनतम Hamming distance चाहिए:

(A) 2
✓ **(B) 3**
(C) 4
(D) 5

✓ **Answer: (B) 3**

Why: Detect e errors: $d \geq e+1 \rightarrow 2+1 = 3$.

क्यों**:** Detect: $d \geq e+1 \rightarrow 3$।

Nukte: Correct 2 errors ke liye $d \geq 2e+1 = 5$.

[Source: Practice — original (textbook-based)] · Hard

Q109. In CRC, if the received codeword is divided by the generator and the remainder is zero, the receiver concludes:

CRC में यदि remainder शून्य हो, तो receiver निष्कर्ष निकालता है:

(A) Error is present / त्रुटि है
✓ **(B) No error detected** / कोई त्रुटि नहीं मिली
(C) Frame is lost / Frame खो गया
(D) ACK is corrupt / ACK ख़राब है

✓ **Answer: (B) No error detected**

Why: Remainder 0 = error nahi mila (probability of undetected error kam).

क्यों: Remainder 0 = त्रुटि नहीं मिली।

Nukte: Non-zero remainder = error.

[Source: Practice — original (textbook-based)] · Medium

Q110. In Selective Repeat ARQ, when a frame is corrupted, the sender retransmits:

Selective Repeat ARQ में frame ख़राब होने पर दोबारा भेजता है:

(A) All outstanding frames / सभी लंबित frames
✓ **(B) Only the corrupted frame** / केवल ख़राब frame
(C) No frame / कोई नहीं
(D) The last frame only / केवल अंतिम frame

✓ **Answer: (B) Only the corrupted frame**

Why: Selective Repeat = sirf galat frame; Go-Back-N = uske aage ke sab.

क्यों: Selective Repeat = सिर्फ़ ख़राब; Go-Back-N = आगे के सब।

Nukte: Go-Back-N sab dobara bhejta hai.

[Source: Practice — original (textbook-based)] · Medium

Q111. Parity check can detect which type of errors?

Parity check कौन-सी errors detect कर सकता है?

(A) Any number of errors / कितनी भी errors
✓ **(B) Odd number of bit errors** / विषम संख्या में bit errors
(C) Only burst errors / केवल burst errors
(D) No errors / कोई नहीं

✓ **Answer: (B) Odd number of bit errors**

Why: Single parity bit odd-bit errors detect karta hai; even ko miss.

क्यों: Parity bit विषम-bit errors detect करती है।

Nukte: CRC burst errors ke liye behtar.

[Source: Practice — original (textbook-based)] · Medium

Q112. Forward Error Correction (FEC) differs from ARQ in that:

Forward Error Correction (FEC), ARQ से भिन्न है क्योंकि:

(A) It needs retransmission / इसे retransmission चाहिए

✓ **(B) The receiver corrects errors without retransmission** / Receiver बिना retransmission errors ठीक करता है

(C) It cannot detect errors / यह errors detect नहीं करता

(D) It works only on LAN / यह सिर्फ़ LAN पर चलता है

✓ **Answer: (B) The receiver corrects errors without retransmission**

Why: FEC receiver par hi error correction karta hai — retransmission nahi.

क्यों: FEC receiver पर ही ठीक करता है।

Nukte: ARQ mein retransmission hota hai.

[Source: Practice — original (textbook-based)] · Medium

Q113. In Go-Back-N ARQ, the receiver's window size is:

Go-Back-N ARQ में receiver की window size होती है:

✓ **(A) 1**

(B) Same as the sender's window / Sender की window के बराबर

(C) 2^(n−1)

(D) Unbounded / असीमित

✓ **Answer: (A) 1**

Why: GBN receiver sirf agla expected frame accept karta hai (window=1); isliye out-of-order frame aaye to use phenk kar pura Go-Back-N retransmission hota hai.

क्यों: GBN में receiver सिर्फ अगला expected frame लेता है (window=1); out-of-order आने पर पूरा Go-Back-N।

Nukte: Selective Repeat mein receiver ki window badi hoti hai — wahi out-of-order frames buffer karta hai.

[Source: Practice — original (textbook-based)] · Medium

Q114. Delaying an acknowledgement so that it can be attached to the next outgoing data frame in the reverse direction is called:

Acknowledgement को रोककर विपरीत दिशा में जाने वाले अगले data frame के साथ जोड़ना कहलाता है:

✓ **(A) Piggybacking**

(B) Broadcasting

(C) Framing

(D) Stuffing

✓ **Answer: (A) Piggybacking**

Why: Piggybacking: bidirectional link par ack ko data frame ke saath chipka do — alag ack frame bachta hai, bandwidth bachti hai.

क्यों: Piggybacking: bidirectional link पर ack को data frame के साथ जोड़ना — bandwidth बचती है।

Nukte: Piggybacking mein ack ke liye timer hota hai, taaki der zyada na ho.

[Source: Practice — original (textbook-based)] · Medium

Q115. In bit stuffing, after how many consecutive 1-bits is a 0 inserted?
Bit stuffing में कितने लगातार 1-bits के बाद 0 जोड़ा जाता है?

(A) 4
✓ **(B) 5**
(C) 6
(D) 8

✓ **Answer: (B) 5**
Why: Paanch 1s ke baad stuffed 0 (HDLC).
क्यों: पाँच 1s के बाद 0 (HDLC)।
Nukte: 6 wale pattern mein flag hota hai.
[Source: Practice — original (textbook-based)] · Hard

Q116. CSMA/CD is used in:
CSMA/CD प्रयोग होता है:

(A) Wireless LAN
✓ **(B) Wired Ethernet**
(C) Satellite links
(D) Fiber only / केवल Fiber

✓ **Answer: (B) Wired Ethernet**
Why: Wired Ethernet mein collision detect possible — CSMA/CD.
क्यों: Wired Ethernet में CSMA/CD।
Nukte: Wireless mein CSMA/CA.
[Source: Practice — original (textbook-based)] · Medium

Q117. In which access method does a station wait for a free time slot before transmitting?
किस access method में station भेजने से पहले ख़ाली time slot की प्रतीक्षा करता है?

(A) Pure ALOHA
✓ **(B) Slotted ALOHA**
(C) 1-persistent CSMA
(D) Polling

✓ **Answer: (B) Slotted ALOHA**
Why: Slotted ALOHA mein sirf slot ke shuruaat par bhej sakte hain.
क्यों: Slotted ALOHA में slot की शुरुआत पर।
Nukte: Pure ALOHA jab mann kare.
[Source: Practice — original (textbook-based)] · Medium

Q118. Token passing is an example of which access method?
Token passing किस access method का उदाहरण है?

(A) Random access
✓ **(B) Controlled access**
(C) Contention
(D) Aloha

✓ **Answer: (B) Controlled access**
Why: Token = controlled access — collision nahi.
क्यों: Token = controlled access।
Nukte: ALOHA/CSMA random hain.
[Source: Practice — original (textbook-based)] · Medium

Q119. Which field of the Ethernet frame is used for error detection?
Ethernet frame का कौन-सा field error detection के लिए है?

(A) Preamble
(B) Type field
✓ **(C) FCS**
(D) MAC address

✓ **Answer: (C) FCS**

Why: FCS (Frame Check Sequence) CRC carry karta hai.
क्यों**:** FCS CRC ले जाता है।
Nukte: Preamble synchronisation ke liye.
[Source: Practice — original (textbook-based)] · Medium

Q120. Gigabit Ethernet over optical fiber is defined by which IEEE standard?
Optical fiber पर Gigabit Ethernet किस IEEE standard में परिभाषित है?

✓ **(A) 802.3z**
(B) 802.3ab
(C) 802.3u
(D) 802.3ae

✓ **Answer: (A) 802.3z**

Why: 802.3z = 1000BASE-X (fiber/STP); 802.3ab = 1000BASE-T (UTP); 802.3u = Fast Ethernet (100 Mbps); 802.3ae = 10 Gbps Ethernet.
क्यों**:** 802.3z = 1000BASE-X (fiber); 802.3ab = 1000BASE-T (UTP); 802.3u = Fast Ethernet; 802.3ae = 10 GbE।
Nukte: Suffix -T = twisted pair, -X = fiber — standard numbers yaad rakho.
[Source: Practice — original (textbook-based)] · Medium

Q121. CSMA/CD (collision detection) can be used only when the link operates in:
CSMA/CD (collision detection) तभी प्रयोग हो सकता है जब link चलती हो:

✓ **(A) Half-duplex mode**
(B) Full-duplex mode
(C) Simplex mode
(D) Both half and full duplex / Half और full duplex दोनों

✓ **Answer: (A) Half-duplex mode**

Why: Collision tabhi detect hoti hai jab ek shared medium par dono taraf se traffic aa sake (half-duplex). Full-duplex (alag pairs) mein collisions hote hi nahi — modern Ethernet mein CSMA/CD off hota hai. Pattern: SSC CGL/IBPS + CCNA.
क्यों**:** Collision shared medium (half-duplex) पर ही होती है। Full-duplex में CSMA/CD बंद। Pattern: SSC CGL + CCNA।
Nukte: Full-duplex = collision-free, isliye minimum frame size ka constraint bhi nahi.
[Source: Practice — original (SSC CGL/IBPS exam-pattern, textbook-based)] · Hard

Q122. Why is the TTL field decremented at each router?
हर router पर TTL field क्यों घटाया जाता है?

(A) To measure speed / Speed नापने के लिए
✓ **(B) To prevent infinite looping of packets** / Packets के अनंत चक्कर रोकने के लिए
(C) To encrypt the packet / Packet encrypt करने के लिए
(D) To compress the packet / Packet compress करने के लिए

✓ **Answer: (B) To prevent infinite looping of packets**

Why: TTL khatam = packet drop (loop prevention).
क्यों**:** TTL ख़त्म = packet drop (loop prevention)।
Nukte: Encryption/compression ka alag kaam.
[Source: Practice — original (textbook-based)] · Medium

Q123. An IP datagram of 1000 bytes (20-byte header) is fragmented on a link with MTU 200 bytes. The number of data bytes in the FIRST fragment is:
1000 bytes का datagram (20-byte header), MTU 200 link पर टूटा। पहले fragment में data bytes होंगे:

✓ **(A) 180**
(B) 200
(C) 980
(D) 160

✓ **Answer: (A) 180**

Why: MTU 200 = 20 header + 180 data; offset 8 ke multiple mein (180 = 22.5×8 → 176 practical; par standard question mein 180 liya jaata hai — GATE style options 180 hi hote hain).
क्यों**:** MTU 200 − 20 = 180 data पहले fragment में।
Nukte: 200 poora frame hai.
[Source: Practice — original (textbook-based)] · Hard

Q124. Which routing protocol uses hop count as its metric with a maximum of 15?
कौन-सा routing protocol hop count metric रखता है (अधिकतम 15)?

(A) OSPF
(B) BGP
✓ **(C) RIP**
(D) ICMP

✓ **Answer: (C) RIP**

Why: RIP: hop count, max 15 (16 = infinity).
क्यों**:** RIP: hop count, अधिकतम 15।
Nukte: OSPF cost use karta hai.
[Source: Practice — original (textbook-based)] · Medium

Q125. Ping utility uses which ICMP messages?
Ping utility किस ICMP messages का प्रयोग करती है?

(A) Redirect
✓ **(B) Echo request and echo reply** / Echo request और echo reply
(C) Time exceeded
(D) Address mask

✓ **Answer: (B) Echo request and echo reply**
Why: Ping = Echo Request (type 8) + Echo Reply (type 0).
क्यों**:** Ping = Echo Request + Echo Reply।
Nukte: Time exceeded traceroute mein.
[Source: Practice — original (textbook-based)] · Medium

Q126. BGP is used for:
BGP प्रयोग होता है:

(A) Routing inside one LAN / एक LAN के अंदर routing के लिए
✓ **(B) Routing between autonomous systems** / Autonomous systems के बीच routing के लिए
(C) Error reporting / Error reporting के लिए
(D) Address resolution / Address resolution के लिए

✓ **Answer: (B) Routing between autonomous systems**
Why: BGP = inter-AS routing (Internet backbone).
क्यों**:** BGP = inter-AS routing।
Nukte: RIP/OSPF intra-AS.
[Source: Practice — original (textbook-based)] · Medium

Q127. Congestion control in the network layer is mainly concerned with:
Network layer का congestion control मुख्यतः संबंधित है:

✓ **(A) Too many packets in the network** / Network में बहुत अधिक packets से
(B) Slow receiver / धीमे receiver से
(C) Broken cables / टूटी cables से
(D) Wrong addresses / ग़लत addresses से

✓ **Answer: (A) Too many packets in the network**
Why: Congestion = network ke andar traffic overload; flow control = slow receiver.
क्यों**:** Congestion = network overload; flow control = धीमा receiver।
Nukte: Slow receiver flow control ki samasya hai.
[Source: Practice — original (textbook-based)] · Medium

Q128. OSPF exchanges routing information directly over IP using IP protocol number:
OSPF routing information IP पर सीधे भेजता है — IP protocol number होता है:

✓ **(A) 89**
(B) 179
(C) 520
(D) 6

✓ **Answer: (A) 89**
Why: OSPF = IP protocol 89 (na TCP na UDP); BGP = TCP 179; RIP = UDP 520; protocol 6 = TCP.
क्यों**:** OSPF = IP protocol 89 (न TCP न UDP); BGP = TCP 179; RIP = UDP 520।
Nukte: Ye teen (89/179/520) worldwide exams ke favourite hain.
[Source: Practice — original (textbook-based)] · Medium

Q129. BGP, the protocol that routes between autonomous systems on the Internet, uses:
BGP — जो Internet पर autonomous systems के बीच routing करता है — प्रयोग करता है:

✓ **(A) TCP port 179**
(B) IP protocol 89
(C) UDP port 520
(D) TCP port 23

✓ **Answer: (A) TCP port 179**
Why: BGP = TCP port 179 — reliable transport chahiye Internet-scale routing ke liye.
क्यों**:** BGP = TCP port 179 — Internet-scale routing के लिए reliable transport।
Nukte: 89 = OSPF; 520 = RIP; 23 = Telnet.
[Source: Practice — original (textbook-based)] · Medium

Q130. The RIP routing protocol uses which transport protocol and port number?
RIP routing protocol कौन-सा transport protocol और port number प्रयोग करता है?

✓ **(A) UDP port 520**
(B) TCP port 520
(C) UDP port 161
(D) TCP port 89

✓ **Answer: (A) UDP port 520**
Why: RIP = UDP 520 (chhote periodic updates, reliable transport ki zaroorat nahi).
क्यों**:** RIP = UDP 520 (छोटे periodic updates)।
Nukte: 161 = SNMP; TCP 89 kuch nahi (OSPF TCP par nahi chalta).
[Source: Practice — original (textbook-based)] · Medium

Q131. A router receives a datagram whose TTL becomes 0 after it decrements the field. The router will:
एक router को datagram मिला जिसका TTL घटाने पर 0 हो गया। router क्या करेगा?

✓ **(A) Discard the datagram and send an ICMP Time Exceeded message to the source** / Datagram नष्ट करके source को ICMP Time Exceeded भेजेगा
(B) Forward it with TTL 0 / TTL 0 के साथ आगे भेजेगा
(C) Send it back to the previous router / पिछले router को वापस भेजेगा
(D) Change TTL to 255 and forward / TTL 255 करके आगे भेजेगा

✓ **Answer: (A) Discard the datagram and send an ICMP Time Exceeded message to the source**
Why: TTL = routing loops rokne ke liye; har hop par −1, zero hote hi packet gira kar ICMP Time Exceeded (type 11) source ko jaata hai — traceroute isi par chalta hai. Pattern: GATE + RPSC Programmer dono ka favourite.
क्यों**:** TTL हर hop पर −1; 0 होते ही packet गिरकर ICMP Time Exceeded (type 11) जाता है — traceroute इसी पर। Pattern: GATE + RPSC Programmer।
Nukte: traceroute = TTL 1, 2, 3... karke Time Exceeded ke source routers ka pata.
[Source: Practice — original (RPSC Programmer/ACP/DD exam-pattern, textbook-based)] · Hard

Q132. In OSPF, on a multiaccess network the Designated Router (DR) is the router with:

OSPF में multiaccess network पर Designated Router (DR) होता है वह router जिसके पास:

✓ **(A) The highest OSPF interface priority (tie → highest router ID)** / सबसे अधिक OSPF interface priority (बराबरी पर → सबसे बड़ा router ID)

(B) The lowest IP address / सबसे कम IP address

(C) The highest bandwidth link / सबसे अधिक bandwidth वाला link

(D) The oldest OSPF process / सबसे पुराना OSPF process

✓ **Answer: (A) The highest OSPF interface priority (tie → highest router ID)**

Why: DR election: pehle highest priority (default 1; 0 = kabhi DR nahi), tie par highest router ID. DR adjacency kam karke flooding reduce karta hai. Pattern: CCNA + RPSC ACP/Programmer level.

क्यों**:** DR election: सर्वोच्च priority (default 1), बराबरी पर सबसे बड़ा router ID। Pattern: CCNA + RPSC ACP/Programmer।

Nukte: OSPF mein 2-way state ke baad DR/BDR election hoti hai.

[Source: Practice — original (RPSC Programmer/ACP/DD exam-pattern, textbook-based)] · Hard

Q133. Two OSPF routers are stuck in the EXSTART/EXCHANGE neighbour state. The most likely cause is:

दो OSPF routers EXSTART/EXCHANGE neighbour state में अटके हैं। सबसे संभावित कारण है:

✓ **(A) An MTU mismatch on the connecting interfaces** / जोड़ने वाले interfaces पर MTU का अंतर

(B) Different OSPF process IDs / भिन्न OSPF process IDs

(C) Different hostnames / भिन्न hostnames

(D) Different IOS/OS versions / भिन्न IOS/OS versions

✓ **Answer: (A) An MTU mismatch on the connecting interfaces**

Why: Database description (DBD) exchange mein MTU mismatch ke routers EXSTART/EXCHANGE se aage nahi badhte. Process IDs locally significant hote hain — match karne ki zaroorat nahi. Pattern: CCNA troubleshooting + RPSC ACP.

क्यों**:** MTU mismatch से routers EXSTART/EXCHANGE में अटकते हैं। Process IDs locally significant होते हैं। Pattern: CCNA + RPSC ACP।

Nukte: OSPF states: Down → Init → 2-Way → ExStart → Exchange → Loading → Full.

[Source: Practice — original (RPSC Programmer/ACP/DD exam-pattern, textbook-based)] · Hard

Q134. The administrative distance of a static route, an EIGRP route and an OSPF route are respectively:

Static route, EIGRP route और OSPF route का administrative distance क्रमशः होता है:

✓ **(A) 1, 90 and 110** / 1, 90 और 110

(B) 0, 110 and 120 / 0, 110 और 120

(C) 1, 110 and 120 / 1, 110 और 120

(D) 10, 90 and 100 / 10, 90 और 100

✓ **Answer: (A) 1, 90 and 110**

Why: AD = route ki 'trustworthiness' (kam = zyada trusted): connected 0, static 1, EIGRP 90, OSPF 110, RIP 120. Pattern: CCNA 200-301 + RPSC ACP/DD.

क्यों**:** AD (कम = अधिक trusted): connected 0, static 1, EIGRP 90, OSPF 110, RIP 120। Pattern: CCNA + RPSC ACP/DD।

Nukte: AD aur metric alag cheez hain — AD protocols ko compare karti hai.

[Source: Practice — original (RPSC Programmer/ACP/DD exam-pattern, textbook-based)] · Hard

Q135. The size of the TCP header (without options) is:
TCP header का size (options बिना) होता है:

(A) 8 bytes
✓ **(B) 20 bytes**
(C) 24 bytes
(D) 60 bytes

✓ **Answer: (B) 20 bytes**
Why: TCP header = 20 bytes; UDP = 8 bytes; IPv4 = 20 bytes.
क्यों**:** TCP = 20 bytes; UDP = 8; IPv4 = 20।
Nukte: 8 bytes UDP ka.
[Source: Practice — original (textbook-based)] · Hard

Q136. During TCP connection termination, after receiving FIN the receiver sends:
TCP connection समाप्ति में FIN पाने के बाद receiver भेजता है:

(A) SYN
✓ **(B) ACK**
(C) RST
(D) PSH

✓ **Answer: (B) ACK**
Why: FIN ka jawab ACK; phir apna FIN bhejta hai.
क्यों**:** FIN का जवाब ACK।
Nukte: SYN establishment ke liye.
[Source: Practice — original (textbook-based)] · Medium

Q137. The size of the UDP header is:
UDP header का size होता है:

✓ **(A) 8 bytes**
(B) 12 bytes
(C) 20 bytes
(D) 64 bytes

✓ **Answer: (A) 8 bytes**
Why: UDP header = 8 bytes (ports + length + checksum).
क्यों**:** UDP = 8 bytes।
Nukte: TCP 20 bytes.
[Source: Practice — original (textbook-based)] · Hard

Q138. In TCP slow start, the congestion window size:
TCP slow start में congestion window size:

(A) Increases by 1 MSS per RTT / प्रति RTT 1 MSS बढ़ती है
✓ **(B) Doubles approximately every RTT** / हर RTT में लगभग दोगुनी होती है
(C) Remains constant / स्थिर रहती है
(D) Decreases every ACK / हर ACK पर घटती है

✓ **Answer: (B) Doubles approximately every RTT**
Why: Slow start exponential growth per RTT.
क्यों**:** Slow start per RTT चरम-वृद्धि।
Nukte: +1 MSS per RTT congestion avoidance hai.
[Source: Practice — original (textbook-based)] · Medium

Q139. TCP is called a byte-stream protocol because:

TCP को byte-stream protocol कहते हैं क्योंकि:

(A) It sends only bytes / यह सिर्फ़ bytes भेजता है

✓ **(B) It does not preserve message boundaries** / यह message सीमाएँ नहीं रखता

(C) It cannot send text / यह text नहीं भेज सकता

(D) It works only with bytes files / यह सिर्फ़ byte files के साथ चलता है

✓ **Answer: (B) It does not preserve message boundaries**

Why: TCP data ko byte stream samajhta hai — message boundaries preserve nahi karta.

क्यों**:** TCP byte stream — सीमाएँ नहीं रखता।

Nukte: UDP message-oriented hai.

[Source: Practice — original (textbook-based)] · Hard

Q140. Which of the following uses UDP?

निम्न में से कौन UDP प्रयोग करता है?

(A) FTP

(B) SMTP

✓ **(C) DNS query**

(D) SSH

✓ **Answer: (C) DNS query**

Why: DNS queries UDP (53) par; FTP/SMTP/SSH TCP.

क्यों**:** DNS query UDP; बाक़ी TCP।

Nukte: Tez chhoti queries UDP.

[Source: Practice — original (textbook-based)] · Medium

Q141. After a TCP connection is closed, the side performing the active close waits in the TIME-WAIT state for:

TCP connection बंद होने के बाद active close करने वाला पक्ष TIME-WAIT state में रहता है:

✓ **(A) 2 × MSL (Maximum Segment Lifetime)**

(B) 1 × RTT

(C) 3 × RTT

(D) One congestion timeout / एक congestion timeout

✓ **Answer: (A) 2 × MSL (Maximum Segment Lifetime)**

Why: 2×MSL wait se purane connection ke delayed duplicate segments khatam ho jaate hain, naya connection safe rehta hai (RFC 793).

क्यों**:** 2×MSL के इंतज़ार से पुराने duplicates खत्म होते हैं (RFC 793)।

Nukte: MSL = maximum time ek segment Internet mein reh sakta hai.

[Source: Practice — original (textbook-based)] · Medium

Q142. The 8-byte UDP header consists of which fields?
8-byte UDP header में कौन-से fields होते हैं?

✓ **(A) Source port, destination port, length, checksum**
(B) Source port, destination port, sequence number, ack number
(C) Source port, destination port, window, checksum
(D) Ports, length, checksum, urgent pointer

✓ **Answer: (A) Source port, destination port, length, checksum**

Why: UDP header = 4 fields × 2 bytes = 8 bytes; sequence/ack/window — sab TCP ke fields hain.

क्यों**:** UDP header = 4 fields × 2 bytes = 8; sequence/ack/window — TCP के fields।

Nukte: UDP ka mantra: 'chhota header, koi guarantee nahi'.

[Source: Practice — original (textbook-based)] · Hard

Q143. TCP provides a byte-stream service. This means that:
TCP byte-stream service देता है — इसका अर्थ है कि:

✓ **(A) Message boundaries are NOT preserved — a record may be split across segments or two records merged into one** / Message boundaries नहीं बनतीं — एक record कई segments में बँट सकता है या दो records एक में मिल सकते हैं
(B) Each send() call arrives as one segment / हर send() एक segment के रूप में पहुँचता है
(C) Bytes arrive in fixed 512-byte blocks / Bytes 512-byte के तय blocks में आते हैं
(D) The receiver delivers bytes in the order they were received, even if reordered on the way / Receiver bytes उसी क्रम में देता है जिसमें प्राप्त हुए, भले रास्ते में बदले हों

✓ **Answer: (A) Message boundaries are NOT preserved — a record may be split across segments or two records merged into one**

Why: TCP sirf reliable ordered BYTE stream deta hai — application ki record boundaries TCP ki zimmedari nahi (isliye application khud length/delimiter bhejti hai). Pattern: RPSC Programmer/ACP/DD + GATE.

क्यों**:** TCP सिर्फ विश्वसनीय क्रमबद्ध BYTE stream देता है — record boundaries की ज़िम्मेदारी नहीं। Pattern: RPSC Programmer/ACP/DD।

Nukte: UDP message-oriented hai — wahan ek datagram = ek message.

[Source: Practice — original (RPSC Programmer/ACP/DD exam-pattern, textbook-based)] · Hard

Q144. The TCP Maximum Segment Size (MSS) is typically the path MTU minus _____ bytes of headers:
TCP Maximum Segment Size (MSS) सामान्यतः path MTU में से _____ bytes के headers घटाने पर मिलती है:

✓ **(A) 40**
(B) 20
(C) 64
(D) 54

✓ **Answer: (A) 40**

Why: MSS = MTU − (20-byte IP header + 20-byte TCP header) = MTU − 40. Ethernet par 1500 − 40 = 1460 bytes. Pattern: RPSC Programmer/ACP/DD + GATE.

क्यों**:** MSS = MTU − (20 IP + 20 TCP) = MTU − 40; Ethernet पर 1460 bytes। Pattern: RPSC Programmer/ACP/DD।

Nukte: MSS sirf TCP data ka size hai — headers alag.

[Source: Practice — original (RPSC Programmer/ACP/DD exam-pattern, textbook-based)] · Hard

Q145. Which protocol is used to retrieve e-mails and keep them on the server as well?
कौन-सा protocol e-mail लाता है और उन्हें server पर भी रखता है?

(A) SMTP
(B) POP3
✓ **(C) IMAP**
(D) FTP

✓ **Answer: (C) IMAP**
Why: IMAP server par messages manage karta hai; POP3 download karke delete.
क्यों**:** IMAP server पर रखता है; POP3 download+delete।
Nukte: SMTP bhejne ke liye.
[Source: Practice — original (textbook-based)] · Medium

Q146. Translation between ASCII and EBCDIC is performed by the:
ASCII और EBCDIC के बीच अनुवाद करती है:

(A) Application layer
✓ **(B) Presentation layer**
(C) Data link layer
(D) Physical layer

✓ **Answer: (B) Presentation layer**
Why: Presentation = translation, encryption, compression.
क्यों**:** Presentation = translation।
Nukte: Application user services deta hai.
[Source: Practice — original (textbook-based)] · Medium

Q147. HTTPS uses which protocol for security?
HTTPS सुरक्षा के लिए कौन-सा protocol प्रयोग करता है?

(A) SSH
✓ **(B) SSL/TLS**
(C) SNMP
(D) ICMP

✓ **Answer: (B) SSL/TLS**
Why: HTTPS = HTTP + SSL/TLS.
क्यों**:** HTTPS = HTTP + SSL/TLS।
Nukte: SSH remote login ka.
[Source: Practice — original (textbook-based)] · Medium

Q148. DHCP is used to:
DHCP प्रयोग होता है:

(A) Resolve names to IPs / नामों को IP में बदलने के लिए
✓ **(B) Assign IP configuration automatically** / IP configuration स्वतः देने के लिए
(C) Encrypt e-mails / E-mail encrypt करने के लिए
(D) Route packets / Packets route करने के लिए

✓ **Answer: (B) Assign IP configuration automatically**
Why: DHCP automatic IP, mask, gateway, DNS deta hai.
क्यों**:** DHCP स्वतः IP देता है।
Nukte: Naam→IP DNS karta hai.
[Source: Practice — original (textbook-based)] · Medium

Q149. In FTP, the control connection uses port number:
FTP में control connection किस port पर चलती है?

(A) 20
✓ **(B) 21**
(C) 25
(D) 80

✓ **Answer: (B) 21**
Why: Control = 21; data = 20.
क्यों**:** Control = 21; data = 20।
Nukte: 20 data connection.
[Source: Practice — original (textbook-based)] · Medium

Q150. Which of the following is a stateless protocol?
निम्न में से कौन stateless protocol है?

(A) FTP
✓ **(B) HTTP**
(C) Telnet
(D) TCP

✓ **Answer: (B) HTTP**
Why: HTTP har request independent — stateless.
क्यों**:** HTTP — stateless।
Nukte: FTP control connection stateful.
[Source: Practice — original (textbook-based)] · Medium

Q151. DHCP uses which pair of UDP ports (server port, client port)?
DHCP कौन-सी UDP ports की जोड़ी प्रयोग करता है (server port, client port)?

✓ **(A) 67 and 68** / 67 और 68
(B) 69 and 70 / 69 और 70
(C) 53 and 54 / 53 और 54
(D) 20 and 21 / 20 और 21

✓ **Answer: (A) 67 and 68**
Why: DHCP server = UDP 67, DHCP client = UDP 68 (DORA process in dono ports par).
क्यों**:** DHCP server = UDP 67, client = UDP 68 (DORA इन्हीं पर)।
Nukte: 69 = TFTP; 53 = DNS; 20/21 = FTP data/control.
[Source: Practice — original (textbook-based)] · Medium

Q152. In the DHCP DORA process, if two DHCP servers send OFFERs, the client usually accepts:
DHCP DORA process में यदि दो DHCP servers OFFER भेजें, तो client सामान्यतः स्वीकार करता है:

✓ **(A) The first OFFER received** / सबसे पहले प्राप्त OFFER
(B) The OFFER with the longest lease / सबसे लंबी lease वाला OFFER
(C) Both and merges them / दोनों, उन्हें मिलाकर
(D) Neither — it uses a static IP / कोई नहीं — static IP लेता है

✓ **Answer: (A) The first OFFER received**

Why: DORA = Discover → Offer → Request → Acknowledge. Client pehli Offer leta hai aur Request broadcast karta hai (baaki servers ko pata chalta hai ki unka offer nahi liya). Pattern: SSC CGL/IBPS + Network+.
क्यों: DORA = Discover → Offer → Request → Ack. Client पहली Offer लेता है। Pattern: SSC CGL + Network+।
Nukte: Do OFFERs ke beech 'race' hoti hai — jo pehle pahunche.
[Source: Practice — original (SSC CGL/IBPS exam-pattern, textbook-based)] · Hard

Topic 7 — Network Security

Q153. Ensuring that data has not been altered in transit relates to which security goal?
यह सुनिश्चित करना कि data रास्ते में बदला नहीं, किस security goal से है?

(A) Confidentiality
✓ **(B) Integrity**
(C) Availability
(D) Authentication

✓ **Answer: (B) Integrity**

Why: Integrity = data badla na gaya ho.
क्यों: Integrity = data अपरिवर्तित।
Nukte: Confidentiality = chhupana.
[Source: Practice — original (textbook-based)] · Hard

Q154. A program that replicates itself across networks without needing a host file is a:
वह program जो बिना host file के network पर खुद की प्रतियाँ बनाता है:

(A) Virus
✓ **(B) Worm**
(C) Trojan
(D) Cookie

✓ **Answer: (B) Worm**

Why: Worm khud spread hota hai; virus ko host file chahiye.
क्यों: Worm खुद फैलता है; virus को host चाहिए।
Nukte: Trojan nakli software ban kar aata hai.
[Source: Practice — original (textbook-based)] · Medium

Q155. Eavesdropping on network traffic is which type of attack?
Network traffic की जासूसी कौन-सा attack है?

✓ **(A) Passive attack**
(B) Active attack
(C) DoS attack
(D) Replay attack

✓ **Answer: (A) Passive attack**

Why: Snoop/sunna = passive (data badalta nahi).
क्यों**:** सुनना = passive।
Nukte: Modification = active.
[Source: Practice — original (textbook-based)] · Medium

Q156. Sending a fake e-mail that appears to come from a bank to steal passwords is called:
बैंक के नाम से नक़ली e-mail भेजकर password चुराना कहलाता है:

✓ **(A) Phishing**
(B) Sniffing
(C) Spoofing
(D) Tunneling

✓ **Answer: (A) Phishing**

Why: Phishing = nakli identity se secrets churana.
क्यों**:** Phishing = धोखे से चोरी।
Nukte: Spoofing nakli address; sniffing sunna.
[Source: Practice — original (textbook-based)] · Medium

Q157. Flooding a server with so many requests that legitimate users cannot use it is a:
Server को इतने requests से भर देना कि असली users न चला सकें:

✓ **(A) DoS attack**
(B) Phishing
(C) Sniffing
(D) Encryption

✓ **Answer: (A) DoS attack**

Why: Denial of Service = availability par hamla.
क्यों**:** DoS = availability पर हमला।
Nukte: Confidentiality par interception hota hai.
[Source: Practice — original (textbook-based)] · Medium

Q158. A SYN flood (denial-of-service) attack exploits which part of network communication?

SYN flood (denial-of-service) attack संचार के किस भाग का दुरुपयोग करता है?

✓ **(A) The TCP three-way handshake** / TCP three-way handshake
(B) The DNS resolution process / DNS resolution process
(C) The ARP cache / ARP cache
(D) The ICMP echo request / ICMP echo request

✓ **Answer: (A) The TCP three-way handshake**

Why: Attacker dheron SYN bhejta hai — server har ek ke liye half-open connection banata hai (SYN-RECV) aur resources khatam ho jaate hain.

क्यों: Attacker ढेरों SYN भेजता है — server की half-open connections से resources खत्म होती हैं।

Nukte: ARP spoofing = cache poisoning; ICMP flood = echo requests — alag attacks.

[Source: Practice — original (textbook-based)] · Medium

Q159. Which algorithm is symmetric?

निम्न में से कौन symmetric algorithm है?

(A) RSA
✓ **(B) AES**
(C) Diffie-Hellman
(D) ECC

✓ **Answer: (B) AES**

Why: AES symmetric; RSA/DH/ECC asymmetric.

क्यों: AES symmetric; बाक़ी asymmetric।

Nukte: Asymmetric mein key pair hota hai.

[Source: Practice — original (textbook-based)] · Medium

Q160. A hash function is mainly used to provide:

Hash function मुख्यतः देता है:

(A) Confidentiality
✓ **(B) Integrity**
(C) Availability
(D) Compression

✓ **Answer: (B) Integrity**

Why: Hash digest se data change detect (integrity).

क्यों: Hash से परिवर्तन पता (integrity)।

Nukte: Encryption confidentiality deta hai.

[Source: Practice — original (textbook-based)] · Medium

Q161. A digital signature provides:
Digital signature देता है:

(A) Confidentiality only / केवल confidentiality
✓ **(B) Authentication and non-repudiation** / Authentication और non-repudiation
(C) Availability
(D) Compression

✓ **Answer: (B) Authentication and non-repudiation**

Why: Private key se sign — pehchaan + inkar na mumkin.

क्यों**:** Private key से sign — पहचान + non-repudiation।

Nukte: Encryption confidentiality.

[Source: Practice — original (textbook-based)] · Medium

Q162. DES is an example of which type of cipher?
DES किस प्रकार का cipher है?

(A) Stream cipher
✓ **(B) Block cipher**
(C) Asymmetric cipher
(D) Hash function

✓ **Answer: (B) Block cipher**

Why: DES/AES block ciphers (64/128-bit blocks); RC4 stream.

क्यों**:** DES/AES block; RC4 stream।

Nukte: RC4 stream cipher hai.

[Source: Practice — original (textbook-based)] · Medium

Q163. The AES symmetric encryption standard supports key sizes of:
AES symmetric encryption standard की key sizes होती हैं:

✓ **(A) 128, 192 and 256 bits** / 128, 192 और 256 bits
(B) 56, 128 and 256 bits / 56, 128 और 256 bits
(C) 1024, 2048 and 4096 bits / 1024, 2048 और 4096 bits
(D) 64, 128 and 192 bits / 64, 128 और 192 bits

✓ **Answer: (A) 128, 192 and 256 bits**

Why: AES (Rijndael): 128/192/256-bit keys, 128-bit blocks; DES = 56-bit key (purana); RSA 1024+ = asymmetric.

क्यों**:** AES: 128/192/256-bit keys, 128-bit blocks; DES = 56-bit; RSA 1024+ = asymmetric।

Nukte: Symmetric = chhoti keys (128-256); asymmetric = badi keys (1024+).

[Source: Practice — original (textbook-based)] · Hard

Q164. In IPSec tunnel mode, which parts of the original packet are encrypted?
IPSec tunnel mode में original packet के कौन-से भाग encrypt होते हैं?

(A) Only the header / केवल header
(B) Only the payload / केवल payload
✓ **(C) The whole original packet** / पूरा original packet
(D) Nothing / कुछ नहीं

✓ **Answer: (C) The whole original packet**

Why: Tunnel mode = poora original packet encrypt + naya header (VPN).

क्यों**:** Tunnel mode = पूरा packet (VPN)।

Nukte: Transport mode mein sirf payload.

[Source: Practice — original (textbook-based)] · Medium

Q165. SSL/TLS works at which layer?
SSL/TLS किस layer पर काम करता है?

(A) Network
✓ **(B) Transport (between transport and application)** / Transport
(C) Physical
(D) Data link

✓ **Answer: (B) Transport (between transport and application)**
Why: SSL/TLS transport ke upar, application ke neeche.
क्यों**:** SSL/TLS transport के ऊपर।
Nukte: IPSec network layer.
[Source: Practice — original (textbook-based)] · Medium

Q166. A VPN creates:
VPN बनाता है:

✓ **(A) A private tunnel over a public network** / Public network पर private tunnel
(B) A new physical cable / नई physical cable
(C) A virus filter / Virus filter
(D) A new protocol / नया protocol

✓ **Answer: (A) A private tunnel over a public network**
Why: VPN = encryption ke saath private tunnel (IPSec/PPTP).
क्यों**:** VPN = encrypted private tunnel।
Nukte: Firewall filter karta hai.
[Source: Practice — original (textbook-based)] · Medium

Q167. Which wireless security standard is the strongest among these?
इनमें से कौन-सी wireless security standard सबसे मज़बूत है?

(A) WEP
(B) WPA
✓ **(C) WPA2**
(D) Open

✓ **Answer: (C) WPA2**
Why: WPA2 (AES) > WPA > WEP > open.
क्यों**:** WPA2 (AES) > WPA > WEP > open।
Nukte: WEP bahut kamzor.
[Source: Practice — original (textbook-based)] · Medium

Q168. PGP (Pretty Good Privacy) is used for:
PGP प्रयोग होता है:

✓ **(A) Secure e-mail** / सुरक्षित e-mail
(B) Routing
(C) Cable testing
(D) Address resolution

✓ **Answer: (A) Secure e-mail**
Why: PGP e-mail encryption + digital signature.
क्यों**:** PGP e-mail encryption + signature।
Nukte: Routing routing protocols का काम।
[Source: Practice — original (textbook-based)] · Medium

Q169. Which IPSec protocol provides encryption (confidentiality) of the packet data?
IPSec का कौन-सा protocol packet data की encryption (confidentiality) देता है?

✓ **(A) ESP**
(B) AH
(C) IKE
(D) HMAC

✓ **Answer: (A) ESP**

Why: ESP (Encapsulating Security Payload) = encryption + integrity + optional authentication; AH = authentication + integrity, encryption nahi.

क्यों**:** ESP = encryption + integrity; AH = authentication + integrity, encryption नहीं।

Nukte: IKE = key exchange ka protocol; HMAC = message authentication code.

[Source: Practice — original (textbook-based)] · Medium

Q170. In which IPSec mode is the original IP header encrypted together with the payload (as used in VPNs)?
IPSec की किस mode में मूल IP header भी payload के साथ encrypt होता है (जैसे VPN में)?

✓ **(A) Tunnel mode**
(B) Transport mode
(C) Transparent mode
(D) Stealth mode

✓ **Answer: (A) Tunnel mode**

Why: Tunnel mode = pura original packet encapsulate + encrypt, naya outer header lagta hai — site-to-site VPNs isi par chalti hain. Transport mode mein sirf payload encrypt hota hai, original header khula.

क्यों**:** Tunnel mode = पूरा packet encapsulate + encrypt, नया outer header; VPN इसी में। Transport mode में सिर्फ payload encrypt।

Nukte: Transport mode = host-to-host security (jaise IPsec ke andar TLS jaisa use).

[Source: Practice — original (textbook-based)] · Medium

Q171. Which wireless security standard uses AES-CCMP encryption (the replacement for the weaker WEP/TKIP)?
कौन-सी wireless security standard AES-CCMP encryption प्रयोग करती है (कमज़ोर WEP/TKIP की जगह)?

✓ **(A) WPA2**
(B) WEP
(C) WPA
(D) Open authentication

✓ **Answer: (A) WPA2**

Why: WPA2 (802.11i) = AES-CCMP encryption; WEP = RC4 (tuta hua); WPA = TKIP (temporary). WPA3 ab SAE + AES-GCMP. Pattern: SSC CGL/IBPS + Network+.

क्यों**:** WPA2 = AES-CCMP; WEP = RC4 (टूटा); WPA = TKIP; WPA3 = SAE। Pattern: SSC CGL + Network+।

Nukte: Wireless security ka evolution: WEP → WPA → WPA2 → WPA3.

[Source: Practice — original (SSC CGL/IBPS exam-pattern, textbook-based)] · Medium